

Mean-Field Global Convergence for Low-Rank Neural Networks

Anonymous Authors¹

Abstract

This work studies the training dynamics of low-rank neural networks with frozen random features in the mean-field regime. When the mean-field dynamics converges, the limit is shown to be a global minimizer; this holds for gradient-based training under standard independent and identically distributed initialization, despite low-rank constraints and nonconvex loss functions. By explicitly incorporating low-rank structure into the network architecture, a tractable mean-field evolution system is derived. Its well-posedness is established, and it is shown that alternating trained and frozen layers preserves the universal approximation property while simplifying the learning dynamics. The analysis identifies a rank-channel feature learning mechanism, in which different low-rank channels specialize to distinct spatial locations and progressively capture higher-frequency components. This mechanism explains both the persistence of global convergence and the emergence of hierarchical frequency learning. Numerical experiments demonstrate that low-rank networks achieve faster convergence and higher accuracy on highly oscillatory targets, while using substantially fewer parameters than full-rank networks.

1. Introduction

Neural networks have achieved remarkable success across a wide range of applications, including computer vision, natural language processing, scientific computing, and data-driven modeling. These advances have reshaped modern machine learning practice and enabled solutions to problems that were previously considered intractable. Despite this empirical success, theoretical understanding of

neural network training remains limited. A central difficulty lies in the highly non-convex nature of the learning landscape. For most architectures and problem settings, the optimization process is poorly understood: it is generally unclear why gradient-based methods succeed in practice, how representations evolve during training, or under what conditions convergence can be guaranteed. As a result, much of current understanding relies on empirical observations rather than rigorous analysis.

Recent progress in mean-field theory has begun to address this gap. In particular, convergence to a global minimizer has been established for full-rank three-layer neural networks without assuming convexity of the loss function (Pham & Nguyen, 2021). This result demonstrates that, for certain architectures, global convergence can occur despite severe non-convexity, providing a rare and valuable theoretical explanation for successful training.

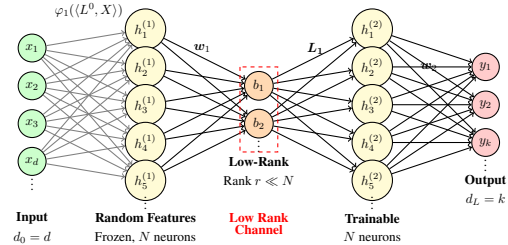
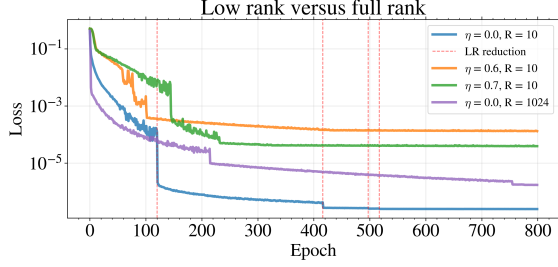


Figure 1. Architecture of a three-layer low-rank random feature network. The first layer consists of frozen random features, the second layer employs a low-rank mixing matrix L of rank $r \ll N$, and the third layer contains trainable weights w_2 . The red dashed box highlights the low-rank bottleneck that reduces dimensionality from N to r channels.

However, full-rank models are heavily over-parameterized. The number of trainable parameters and the associated computational cost can be far larger than what is effectively needed to represent the target function. From both computational and modeling perspectives, it is natural to seek more structured and efficient alternatives. Low-rank neural networks offer such a possibility by reducing parameter redundancy and computation while retaining expressive power. As illustrated in Figure 2, low-rank models achieve comparable or better training loss with 99% fewer parameters than full-rank networks.

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.



rank $r=10$ vs full-rank ($r=1024$) on $y = \cos(2\pi x)$, $x \in [-1, 1]$, 3 layers, width 1024, $n=5000$, batch=4, SGD lr 0.01 with red-bar decay. (momentum $\eta \in \{0, 0.3, 0.6, 0.7\}$) and one full-rank ($\eta=0$). Red bars: SGD lr reductions for $\eta=0$.

Figure 2. Low-rank (rank $r=10$) vs full-rank ($r=1024$) on $y = \cos(2\pi x)$, $x \in [-1, 1]$, 3 layers, width 1024, $n=5000$, batch=4, SGD lr 0.01 with red-bar decay. (momentum $\eta \in \{0, 0.3, 0.6, 0.7\}$) and one full-rank ($\eta=0$). Red bars: SGD lr reductions for $\eta=0$. Low-rank uses 99% fewer parameters.

This leads to a fundamental question: does global convergence persist under low-rank constraints? More specifically, when full-rank weight matrices are replaced by low-rank factorizations, can gradient-based training still be guaranteed to converge to a global minimizer, or is full rank essential for existing convergence results?

Low-rank networks factorize weight matrices into products of tall, skinny matrices (for example, $W = LR^T$ with rank $r \ll \min\{n, m\}$), substantially reducing the number of trainable parameters. Yet, from a theoretical perspective, it remains unclear whether the favorable optimization properties observed in full-rank mean-field analyses extend to this low-rank setting.

These considerations motivate a systematic theoretical investigation of low-rank neural networks from the perspectives of optimization and representation learning. The central objective is to determine whether the favorable global convergence properties recently established for full-rank mean-field models persist under low-rank constraints, and, if so, how the resulting representations are structured at convergence. This question is resolved in the present work, as summarized by the following informal theorem.

Theorem 1.1 (Convergence to a global minimizer for any depth with i.i.d. initialization (informal)). *For any depth $L \geq 2$, low-rank random feature networks: if the weights in all layers $(w_1, \dots, w_{L-1})$ converge as $t \rightarrow \infty$, then the limit is a global minimizer of the population loss.*

Theorem 1.1 is an informal statement of the main convergence result proved rigorously in Theorem 3.2. Beyond global convergence, the analysis further establishes a hierarchical frequency learning phenomenon for low-rank neural networks, revealing how learned representations are organized across different low-rank channels. The main contributions of this work are summarized as follows.

- **Global convergence under low-rank constraints** (Theorem 3.2). It is shown that when the dynamics of low-rank random feature neural networks converges, it converges only to global minimizers of the population loss, for any depth $L \geq 2$ under standard independent and identically distributed initialization. Unlike previous full-rank analyses, the result requires no special or ad-hoc initialization.
- **Mean-field feature learning** (Theorem 4.1). A theoretical characterization of feature learning mechanism is established. A rigorously analyzed toy model demonstrates that each low-rank channel learns a spatially localized feature at a *distinct location*, while Lower-frequency components are captured first, followed by progressively higher-frequency components.

Numerical experiments confirm that this theoretical structure translates into practical benefits. On highly oscillatory function, low-rank networks achieve substantially faster convergence and significantly higher accuracy, reaching $\text{MSE} \sim 10^{-6}$, compared to approximately 10^{-5} for full rank networks. **At the same time, the number of trainable parameters is reduced by 95%–99%** These results suggest a fundamental insight into the loss landscape: appropriately structured low-rank parameterizations can improve optimization behavior, rather than merely restricting expressivity.

A key theoretical advance of this work is the elimination of the ad-hoc initialization assumptions that restrict existing full-rank multi-layer convergence results to depth $L = 3$ (Nguyen & Pham, 2023). In the low-rank random feature setting, frozen random features ensure that $\text{supp}(L^0(C_1)) = \mathbb{R}^d$ is preserved throughout training. As a consequence, standard independent and identically distributed initialization suffices, and when the dynamics converges, it is to a global minimizer for *arbitrary depth* $L \geq 2$.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 presents the main theoretical results and proof ideas. Section 4 studies hierarchical frequency learning and channel specialization. Section 5 reports numerical experiments, and Section 6 concludes.

2. Related Work

2.1. Mean-Field Theory for Neural Networks

Mean-field theory rigorously analyzes neural network training dynamics in the large-width limit (typically $N \geq 1000$ neurons per layer). In this regime, the empirical distribution of parameters evolves deterministically, leading to powerful tools for understanding optimization and gen-

eralization. Foundational results by (Mei et al., 2018), (?), (Rotskoff & Vanden-Eijnden, 2019), and (Wei et al., 2019) established global convergence for two-layer networks under convex loss assumptions—crucially leveraging convexity to guarantee that all stationary points are global minimizers.

Progress beyond the convex case has accelerated. (Pham & Nguyen, 2021) proved global convergence for three-layer networks without assuming convex loss functions. Mean-field analyses of residual networks (Lu et al., 2020) and multilayer networks (Sirignano & Spiliopoulos, 2021) further clarify when global optima arise from stationary points, but these results so far apply only to full-rank networks. Whether such convergence persists in computationally efficient, low-rank architectures remained open.

We address this gap: we avoid Riemannian gradient descent (RGD) by training only the right factor R , keeping the left factor L frozen as random features. by leveraging frozen random features, we prove that channel feature learning (in which r independent channels capture hierarchical frequency structure) leads to both global convergence and practical representational learning.

The present analysis is closely related to the mean-field framework of Chizat and Bach (?), who established a global optimality result (Theorem 2) for two-layer networks. Specifically, if the support of the initial distribution spans all directions in $\mathbb{R}^{d+1}$ and the feature map Ψ is positively 2-homogeneous, then any weak limit of the Wasserstein gradient flow must correspond to a global minimizer of the population risk. The results developed here extend this principle to *low-rank, multi-layer* neural networks.

The analysis adapts mean-field techniques for random feature models, building on the convergence theory developed in (Pham & Nguyen, 2021; Nguyen & Pham, 2023). The introduction of frozen random features allows the removal of the ad-hoc initialization conditions imposed in (Nguyen & Pham, 2023); see Remark 3.3 for a detailed discussion. For a comprehensive account of the full-rank setting, the reader is referred to (Nguyen & Pham, 2023).

2.2. Low-Rank Neural Networks

Our RF-LR architecture is based on (?). Most theoretical work on low-rank networks focuses on expressivity and approximation power, while their training dynamics—especially regarding convergence—have received less attention. (?) study end-to-end low-rank training, often using reparameterization gradient descent (RGD); in contrast, we train only the post-activation parameters, keeping the other factor as frozen random features. This architectural choice enables us to prove that, when the dynamics converges, the limit is a global minimizer for low-rank

networks, and to elucidate the mechanism by which channel feature learning provides both theoretical and empirical benefits: each channel evolves largely independently, specializing to distinct spatial and frequency patterns.

2.3. Random Features and Neural Tangent Kernel

Random features provide a bridge between neural networks and kernel methods. In the mean-field width limit, fixing the first layer as random features connects the network to kernel methods governed by the neural tangent kernel (NTK) (Jacot et al., 2018), offering tractable analyses of optimization and generalization. Our architecture embraces this by freezing the first-layer weights, so that low-rank mixing layers can learn task-specific representations atop a fixed random feature basis. Recent work establishes that low-rank structure suffices for the MLP NTK (Anonymus, 2026), while our results extend this to mean-field dynamics and global convergence. Together, these advances show that, both in NTK and mean-field regimes, low-rank networks can match the optimization guarantees of their full-rank counterparts.

3. Main Results and Proof Ideas

3.1. Low-Rank Random Feature Architecture

We consider a low-rank random feature (RF-LR) network architecture. Let $h^{(0)}(x) = x \in \mathbb{R}^{d_0}$. For layers $\ell = 1, \dots, L$:

$$h^{(\ell)}(x) = \frac{1}{n_\ell} \sum_{j=1}^{n_\ell} w_j^{(\ell)} \varphi_\ell \left(L_j^{(\ell)\top} h^{(\ell-1)}(x) + b_j^{(\ell)} \right) + c_\ell, \quad (1)$$

where n_ℓ is the width of layer ℓ , $w_j^{(\ell)} \in \mathbb{R}^{d_\ell}$ are trainable coefficient vectors, $L_j^{(\ell)} \in \mathbb{R}^{d_{\ell-1}}$ and $b_j^{(\ell)} \in \mathbb{R}$ are frozen random features (i.i.d. uniform or Gaussian), and c is a trainable scalar output bias. The $1/n_\ell$ scaling ensures a well-defined mean-field width limit. In the mean-field width limit (typically $N \geq 1000$ neurons per layer), where N denotes the width of each layer, under the mean-field parameterization with i.i.d. initialization (uniform or Gaussian), the empirical distribution of weights converges to a deterministic measure-valued evolution.

Training policy: $w^{(\ell)}$ and c_ℓ are trained; $L^{(\ell)}$ and $b^{(\ell)}$ are frozen random draws. In the mean-field formulation, biases $b_j^{(\ell)}$ are encoded by augmenting the input data with a constant component (adding 1 to the input vector).

3.2. Mean-Field Forward Equations

We use the *neuronal embedding* framework of (Nguyen & Pham, 2023): each neuron is indexed by a label C lying in a probability space (countable or uncountable), which

we specify when sampling or constructing the network. In the finite-width case, C corresponds to discrete neuron indices; in the mean-field limit, C runs over the support of a measure, and expectations $\mathbb{E}_C[\cdot]$ replace empirical averages over neurons.

Definition. We define the mean-field forward pass layer by layer. The *frozen* first-layer feature map is $L^0(c_1) \in \mathbb{R}^{d_0}$; we specify the law of $L^0(C_1)$ when sampling the network. In layers $i \geq 2$, the low-rank structure is given by a *frozen* mixing matrix L (entries $L_{c_2,k}$, or $L_{c_i,k}^{(i)}$ for intermediate layers), also fixed random features drawn at initialization; we call it *mixing* in those layers. All other layer-wise quantities are defined exactly as follows.

- **Layer $i = 0$ (input):** $h^0(X) = X \in \mathbb{R}^{d_0}$. No neuronal index.

- **Layer $i = 1$ (first hidden):** Neuronal index $C_1 \in C_1$. Frozen feature map $L^0(c_1) \in \mathbb{R}^{d_0}$. Channel-wise partial functions f_k , $k \in \{1, \dots, r\}$:

$$f_k(t; X, W) = \mathbb{E}_{C_1} \left[\underbrace{w_1(t, C_1, k)}_{w_j^{(1)}(\text{train.})} \underbrace{\varphi_1(\langle L^0(C_1), X \rangle)}_{a_1(c_1; X), \text{ pre-act.}} \right],$$

with activation $\varphi_1(a_1)$.

- **Layer $i = 2$ (second hidden):** Neuronal index $C_2 \in C_2$. Pre-activation and activation:

$$\underbrace{H_2(t, c_2; X, W)}_{\text{pre-act.}} = \sum_{k=1}^r \underbrace{L_{c_2,k}}_{\text{mixing}} \underbrace{f_k(t; X, W)}_{\text{channel}}, \quad \underbrace{\varphi_2(H_2)}_{\text{activation}}.$$

- **Layer $i = 3$ (output):** Network output:

$$\hat{y}(X; W(t)) = \mathbb{E}_{C_2} \left[\underbrace{w_2(t, C_2)}_{w_j^{(3)}(\text{train.})} \underbrace{\varphi_2(H_2(t, C_2; X, W(t)))}_{\text{activation}} \right]$$

- **Layers $i \in \{2, \dots, L-1\}$ (intermediate, $L > 3$):** Pre-activation

$$H_i(t, c_i; X, W) = \sum_{k=1}^r \underbrace{L_{c_i,k}^{(i)}}_{\text{mixing}} \underbrace{f_k^{(i)}(t; X, W)}_{\text{channel, layer } i},$$

where

$$f_k^{(i)}(t; X, W) = \mathbb{E}_{C_{i-1}} [w_{i-1}(t, C_{i-1}, k) \varphi_{i-1}(H_{i-1}(t, C_{i-1}; X, W))],$$

with H_{i-1} the pre-activation at layer $i-1$.

3.3. Mean-Field Backward Equations (ODEs)

The mean-field ODE system (backward equations) for the weights takes the form, for all layers $i = 1, \dots, L-1$:

$$\begin{aligned} \partial_t w_1(t, c_1, k) &= -\xi_1(t) \mathbb{E}_Z [d_L \varphi_1(\langle L^0, X \rangle) B_k^{(2)}], \\ \partial_t w_i(t, c_i) &= -\xi_i(t) \mathbb{E}_Z [D_i \varphi_i(H_i(t, c_i; X, W))], \end{aligned} \quad (2)$$

where $\mathbb{E}_Z[\cdot]$ denotes expectation over $Z = (X, Y)$; $d_L(Z; W(t)) = \partial_{\hat{y}} \mathcal{L}(Y, \hat{y}(X; W(t)))$ is the loss derivative; $\xi_i(t) \geq 0$, $i = 1, \dots, L-1$, are the learning-rate schedules. We have $D_{L-1} = d_L$ and D_i for $i < L-1$ is the back-propagated derivative from the layers above. The gradient at each layer is (upstream signal) $\times$ (local derivative) w.r.t. w ; the channel-wise partial functions and back-propagated signals are

$$\begin{aligned} f_k^{(1)}(t; X, W) &= \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), X \rangle)], \quad (f_k := f_k^{(1)}), \\ f_k^{(i)}(t; X, W) &= \mathbb{E}_{C_{i-1}} [w_{i-1}(t, C_{i-1}, k) \varphi_{i-1}(H_{i-1}(t, C_{i-1}; X, W))], \\ B_k^{(\ell)}(t; X, W) &= \mathbb{E}_{C_\ell} [L_{C_\ell, k}^{(\ell)} \varphi'_\ell(H_\ell(t, C_\ell; X, W)) w_\ell(t, C_\ell)], \quad \ell = 2, \dots, L \end{aligned}$$

$L_{c_2,k}^{(2)} = L_{c_2,k}$. The channel- k backpropagated signal $B_k^{(\ell)}$ aggregates gradient information from layer ℓ through the mixing $L^{(\ell)}$ and φ'_ℓ ; for ReLU, $\varphi'_2(u) = \mathbb{1}\{u > 0\}$ (ReLU gate). In the 3-layer case, $B_k := B_k^{(2)}$.

3.4. Assumptions

The applicability of our results depends crucially on the following assumptions, which we state explicitly in Appendix B. These assumptions are the core of the theoretical framework and determine when convergence to a global minimizer can be guaranteed and are **satisfied by standard MLP architectures**

The main assumptions are: *Bounded Activations and Mixing* (Assumption B.1), *Sub-Gaussian Initialization* (Assumption B.2), *Data Distribution and Loss Regularity* (Assumption B.3), *Diversity of Random Features* (Assumption B.4), *Non-Degeneracy* (Assumption B.5), and *Training convergence to limit point* (Assumption B.7). The *Non-Degeneracy* assumption requires that the initial loss is better than the trivial zero predictor, ensuring the network learns a non-trivial solution. The *Convergence to Limit Point* assumption states that training reaches a limit point, which is a natural condition for analyzing convergence.

These assumptions are satisfied by ReLU/SwiGLU networks on bounded data with standard initialization (Gaussian/Xavier).

3.5. Mean-Field ODEs are well posed

We establish well-posedness of the mean-field ODE system (2) by adapting the proof from (Nguyen & Pham, 2023) to account for the low-rank structure.

Theorem 3.1 (Well-posedness of mean-field ODEs). *Under Bounded Activations and Mixing (Assumption B.1), Sub-Gaussian Initialization (Assumption B.2), Data Distribution and Loss Regularity (Assumption B.3), and Diversity of Random Features (Assumption B.4), there exists a unique solution to the mean-field ODE system (2) on $t \in [0, \infty)$.*

Proof sketch. We adapt and introduce weight-space Orlicz sub-gaussian norms accounting for the r channels ($\max_{1 \leq k \leq r}$ over channel index k , see Appendix C.2), establish sub-Gaussian a priori bounds with r -factors, then show the solution operator F is contractive in these spaces; Banach fixed point yields existence and uniqueness. Details in Appendix C. $\square$

3.6. Global convergence

Frozen random features with full support $\text{supp}(L^0(C_1)) = \mathbb{R}^d$ ensure that $\{\text{ReLU}(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\}$ has dense span in $L^2(\mathcal{P}_X)$, since $\text{ReLU}(af + b)$ for random a, b always forms a dense span. This property is maintained throughout training because $L^0(C_1)$ are frozen. For well-posedness (Theorem 3.1), each $H_\ell(t, c_\ell; X, W)$ bi-Lipschitz in W is sufficient and equivalent to φ_ℓ Lipschitz; it is the only thing that matters. After defining norms, we update $K_0(t)$ with a factor $(1 + rK)^{1/2}$; the solution operator F , a priori bounds, and the contraction argument then proceed as in the full-rank framework (Nguyen & Pham, 2023).

Theorem 3.2 (RF-LR training only converges to global minimizers). *Under all assumptions from Bounded Activations and Mixing (Assumption B.1) through Convergence to Limit Point (Assumption B.7), if the mean-field dynamics for low-rank random feature networks converges, then it is to a global minimizer of the loss function: $\lim_{t \rightarrow \infty} W(t) = W^*$ where W^* minimizes the population loss. For any depth $L \geq 2$, this holds with standard i.i.d initialization.*

Equivalently, any limit point of the mean-field dynamics is a global minimizer of the population loss.

High-level proof idea (after (Nguyen & Pham, 2023), Sec. 6.2.1). At a limit point $\bar{W}$, the gradient-flow ODE has zero time derivative. For the top layer (and, by back-prop, at each layer), this yields $\mathbb{E}_Z[\text{upstream} \times \text{local}] = 0$ over the support of the layer’s neuron measure. The crucial step: if the first-layer features $\{\varphi_1(\langle \theta, \cdot \rangle) : \theta \in \text{supp}(\text{feature measure})\}$ have dense span in $L^2(\mathcal{P}_X)$, then one deduces $\mathbb{E}_Z[\partial_{\hat{y}} \mathcal{L}(Y, \hat{y}(X; \bar{W})) \mid X = x] = 0$ for $\mathcal{P}_X$ -a.e. x , hence $\hat{y}(x; \bar{W})$ satisfies the first-order condition for a global minimizer. There is no critical reliance on convexity.

In (Nguyen & Pham, 2023), the first-layer weights w_1 are

trained, so $\text{supp}(\bar{w}_1(U_1)) = \mathbb{R}^d$ at the limit may fail (e.g. sparse or spiky solutions). Their proof therefore relies on a homotopy argument to show that $\text{supp}(w_1(t, U_1)) = \mathbb{R}^d$ is preserved for all finite $t \geq 0$, so the limit inherits enough structure. **Our trick:** we freeze the first-layer feature map $L^0(C_1)$. The pre-activation is $\langle L^0(C_1), X \rangle$; L^0 is not trained, so $\text{supp}(L^0(C_1)) = \mathbb{R}^d$ holds at init and at all t , including the limit. Thus $\{\varphi_1(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\}$ has dense span in $L^2(\mathcal{P}_X)$ throughout training. We can therefore apply the same global-optimality argument as in (Nguyen & Pham, 2023) without the homotopy step: at a limit point, zero derivative plus dense span implies $\mathbb{E}_Z[\partial_{\hat{y}} \mathcal{L} \mid X = x] = 0$ a.e., hence the limit is a global minimizer. The low-rank structure ($H_2 = \sum_k L_{c_2, k} f_k$, etc.) only changes the form of the upstream and local terms; the gradient-flow logic and the use of dense span at the first layer are unchanged. For $L \geq 3$, the same layer-by-layer argument applies once the first layer supplies dense span via frozen L^0 . Full-rank arguments extend without ad-hoc init (Remark 3.3).

Remark 3.3 (Avoiding ad-hoc initialization). Full-rank multilayer nets ($L \geq 3$) need *ad-hoc initialization* in that framework: otherwise intermediate layers collapse and diversity fails (training can collapse the required full support). We avoid this: frozen features keep $\text{supp}(L^0(C_1)) = \mathbb{R}^d$, so standard i.i.d. initialization suffices and, when the dynamics converges, it is to a global minimizer for any L .

3.7. Quantitative Guarantees

We provide a quantitative approximation theorem that bounds the error between finite-width networks and the mean-field limit. The detailed proof are provided in Appendix H; here we summarize the key result.

Theorem 3.4 (Finite-width approximation error bound). *Given a family Init of initialization laws and a tuple $\{n_1, n_2\}$ that is in the index set of Init , perform the coupling procedure for the low-rank architecture as described in Section 3.1. Fix a terminal time $T \in \mathbb{N}_{\geq 0}$. Under Assumptions B.1, H.1 (see Appendix H), and the low-rank structure with mixing matrix L satisfying $\|L\|_{\infty, 1} \leq rK$, for $\epsilon \leq 1$, we have with probability at least $1 - 2\delta$,*

$$\mathcal{D}_T(W, \mathbf{W}) \leq C_{\text{exp}} \cdot C_{\text{width}} \cdot C_{\text{log}},$$

where $C_{\text{exp}} = e^{K_T(1+rK)}$, $C_{\text{width}} = 1/\sqrt{n_{\min}} + \sqrt{\epsilon}$, $C_{\text{log}} = \sqrt{\log(3(T+1)n_{\max}^2/\delta + e)}$, with $n_{\min} = \min\{n_1, n_2\}$, $n_{\max} = \max\{n_1, n_2\}$, $K_T = K(1 + T^K)$, and the factor $(1 + rK)$ accounts for the low-rank structure through $\|L\|_{\infty, 1} \leq rK$.

The exponential factor $e^{K_T(1+rK)}$ reflects the channel feature learning structure of low-rank networks: the r channels evolve independently through the mixing matrix L , and the worst-case time complexity becomes exponential in rK

when all channels must be learned simultaneously. However, channel feature learning enables a choice of which channels to activate, and in practice, only a subset of channels are needed, avoiding the worst-case exponential scaling.

The theorem links $\mathbf{W}(\lfloor t/\epsilon \rfloor)$ (discrete finite-width) to $W(t)$ (mean-field limit): error $O(1/\sqrt{n_{\min}} + \sqrt{\epsilon})$, independent of d . Versus full-rank (Nguyen & Pham, 2023), the exponential is $e^{K_T(1+rK)}$ instead of e^{K_T} because the r channels evolve through L ($\|L\|_{\infty,1} \leq rK$); effective capacity is constrained by r . Proof in Appendix H.

4. Feature Learning

The low-rank structure enables channel feature learning where different channels learn different spatial-frequency features. In particular, each channel learns a spike at a *different spatial value* (localization): e.g. channel k dominates near some x_k while others remain small there. Empirically, channels also separate by *frequency*—lower frequencies are captured first, higher ones progressively—so that the r channels jointly provide both spatial localization and frequency separation. The following lemma provides a rigorous characterization of how channels establish and maintain dominance, explaining the feature learning in practice.

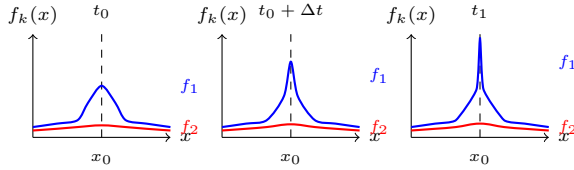


Figure 3. Channel spike learning (Theorem 4.1). Functions $f_1(x)$ (blue) and $f_2(x)$ (red) at t_0 , $t_0 + \Delta t$, and t_1 . f_1 has a spike at x_0 that becomes more pronounced over time, while f_2 has no spike.

4.1. Mechanism of Spike Learning

The log-ratio growth in Theorem 4.1 arises from a self-reinforcing dominance loop. The evolution $\partial_t f_k(t, x) = -\xi_1(t) \mathbb{E}_{Z=(X,Y)}[d_L(Z; W(t)) B_k(t; X) K_{\mu_0}(x, X)]$ is an integral over the data distribution, where $k \in \{1, \dots, r\}$ is the channel index and $K_{\mu_0}(x, X)$ is the kernel. When channel k dominates at x_0 ,

$$H_2 = \sum_{j=1}^r L_{c_2,j} f_j = \underbrace{L_{c_2,k} f_k}_{\text{large}} + \underbrace{\sum_{j \neq k} L_{c_2,j} f_j}_{\text{small}},$$

so the pre-activation and hence the gate $\mathbb{1}\{H_2 > 0\}$ are effectively driven by that channel. The backpropagated signal $B_k(t; X) = \mathbb{E}_{C_2}[L_{C_2,k} w_2(t, C_2) \mathbb{1}\{H_2(t, C_2; X) > 0\}]$ is a gated mixture over the second layer. The precise

signs of $L_{c_2,k}$ or $w_2(t, c_2)$ need not matter: once a channel dominates, the ReLU gate selects a half-space (e.g. $\mathbb{1}\{L_{C_2,j} > 0\}$), w_2 increases on that set via the mean-field w_2 -ODE, and the dynamics create an *emergent* sign-coherence rather than requiring it a priori. If the activation (gate) is correlated with a channel, that correlation *amplifies*: the channel gets a larger B_k , hence a larger $\partial_t f_k$, so the channel and the gate become even more correlated. Under sign-coherence ($B_{k,1}$ has the same sign as f_k), the integral contribution from the dominant channel is larger, yielding a single self-reinforcing loop: larger dominance $\rightarrow$ larger $B_k \rightarrow$ larger $\partial_t f_k \rightarrow$ further amplified dominance. These mechanisms are proven in the two-sided step dataset toy model (Theorem 4.1); see Appendix F.

4.2. Toy model with provable spike feature learning

The full mean-field system for discrete regression in Spatial-Fourier space is infinite dimensional; the two-point ($m = 2$) dynamics nonetheless capture the essential mechanism for feature learning. We state and prove the main result for the *two-sided step* toy model: $m = 2$ support points $x^{(1)} = x_0 = -\delta$, $x^{(2)} = x_1 = +\delta$ ($\delta > 0$) with $y^{(1)} = +A$, $y^{(2)} = -A$. The target $y(x)$ is supported on $\{x_0, x_1\}$ with values $\pm A$ (Figure 4). For finitely-supported data, the w_1 -evolution is linear in feature space once residuals and backprop signals are fixed; one obtains closed-form f_k as a superposition of kernel bumps. Define $d_p(t) = d_L((x^{(p)}, y^{(p)}); W(t))$, $B_{k,p}(t) = B_k(t; x^{(p)})$, and

$$\Gamma_{k,p}(t) \equiv \int_0^t \xi_1(s) d_p(s) B_{k,p}(s) ds.$$

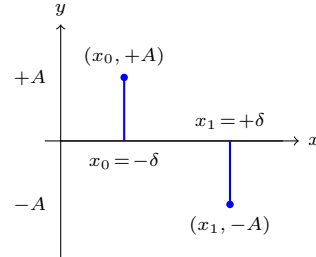


Figure 4. Two-sided step: $x_0 = -\delta$, $x_1 = +\delta$ with $y(x_0) = +A$, $y(x_1) = -A$.

Then

$$f_k(t, x) = f_k(0, x) - \Gamma_{k,1}(t) K_{\mu_0}(x, x_0) - \Gamma_{k,2}(t) K_{\mu_0}(x, x_1). \quad (3)$$

The spike shape is determined by K_{μ_0} ; all learning dynamics reduce to the *scalar* coefficients $\Gamma_{k,1}(t)$, $\Gamma_{k,2}(t)$. The kernel K_{μ_0} (NNGP, (Jacot et al., 2018)) is Laplace-like: $K_{\mu_0}(x_p, x_p) = K_0 > 0$ for $p \in \{0, 1\}$ (same by symmetry), and $|K_{\mu_0}(x_0, x_1)| = |K_{\mu_0}(-\delta, +\delta)| \leq C e^{-c\delta}$ for some $C, c > 0$, so the off-diagonal is exponentially small in δ .

The full evolution has *two terms* (local plus non-local):

$$\partial_t f_k(t, x_p) = -\xi_1(t) K_{\mu_0}(x_p, x_p) d_p(t) B_{k,p+1}(t) + E_p(t), \quad (4)$$

where the non-local remainder $E_p(t)$ satisfies $|E_p(t)| \leq C' e^{-c\delta}$ for some $C', c > 0$. We state the theorem for x_0 only; the result is symmetrical at x_1 (Appendix E).

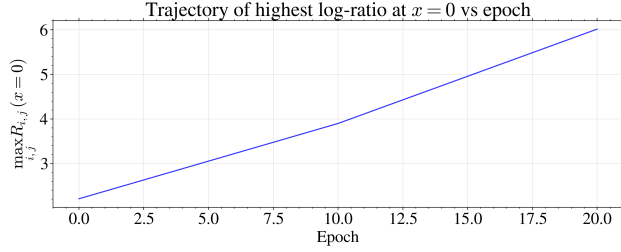


Figure 5. Trajectory of $\max_{i,j} R_{i,j}(x=0)$ vs. epoch (layer-3 channels). Channel specialization at $x=0$ as in Theorem 4.1. Setup: $n=1024$, $r=15$, $\cos(8\pi x)$, 20 epochs.

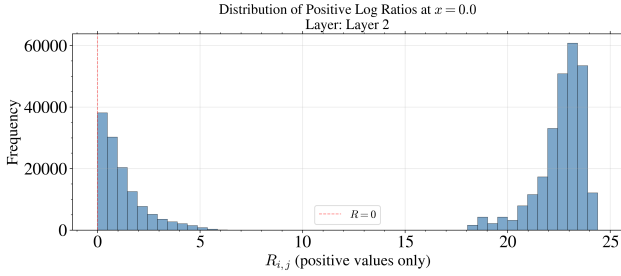


Figure 6. Log-ratios $R_{i,j}$ at $x=0$ over training (all activation pairs; layer-2 H_k).

Theorem 4.1 (Two-sided step: log-ratio growth at x_0). *Assume: the two-sided step setting above with $r=2$ channels and a 3-layer RF-LR; the dynamics are (3) and the full evolution (4).*

Definition (log-ratio criterion). Let $I \subseteq [0, \infty)$. For $p \in \{0, 1\}$, $d_p(t)$ is the residual at x_p and $B_{k,p+1}(t) = B_k(t; x^{(p+1)})$. (Here $x^{(p+1)} = x_p$ since $x^{(1)} = x_0$ and $x^{(2)} = x_1$, so $B_{k,p+1}$ is the backprop signal at x_p .) Define

$$R_{12}(t, x_0) = \log \frac{|f_1(t, x_0)|}{|f_2(t, x_0)|}.$$

Suppose (at x_0): $-d_0(t) \geq 0$ on I ; $B_{1,1}(t)$ has the same sign as $f_1(t, x_0)$; and there exists $\rho_0 \in [0, 1)$ such that $|B_{2,1}(t)| \leq \rho_0 \frac{|f_2(t, x_0)|}{|f_1(t, x_0)|} |B_{1,1}(t)|$ for all $t \in I$.

Then: For $t \in I$,

$$\partial_t R_{12}(t, x_0) = (1 - \rho_0) \xi_1(t) K_{\mu_0}(x_0, x_0) (-d_0(t)) \frac{|B_{1,1}(t)|}{|f_1(t, x_0)|} + \varepsilon_0(t),$$

with $|\varepsilon_0(t)| \leq C'' e^{-c\delta}$ (from E_p). Hence $\partial_t R_{12}(t, x_0) \geq 0$ whenever the leading term dominates $|\varepsilon_0|$; strict dominance of channel 1 at x_0 cannot be lost on I and is amplified whenever $|B_{1,1}|$ is not too small. The result is symmetrical at x_1 (channel 2 dominates there).

Proof sketch. The proof is deferred to Appendix F and only uses Equation (4) and the log-ratio calculus at x_0 . $\square$

5. Numerical Results

Our experiments focus solely on optimization in 1-dimensional data to use large sample sizes to ensure that convergence of $\mathbb{E}_Z[\cdot]$ is achieved. Hence, optimization dynamics are not affected by overfitting or sample complexity.

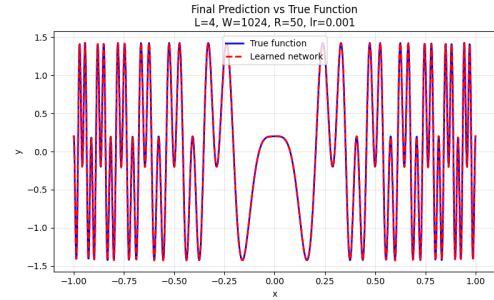


Figure 7. Final prediction vs. target $f(x) = \cos(36\pi x^2) - 0.8 \cos(12\pi x^2)$. 4-layer, $n=1024$, $r=50$, $N=1000$, RF-LR MSEs are within 10^{-8} to $\sim 10^{-4}$

Table 1. Hyperparameters for all figures in this section. Symmetry runs (Figs. 8, 9, 8) and Fig. 7: Adam (Kingma & Ba, 2015), batch 100, lr 0.001, $\gamma=0.9$ every 100 steps; test 4936 for symmetry. Log-ratio (Figs. 5, 6): SGD, lr 0.01, batch 160. Target (f_1, f_2) denotes $f(x) = \cos(f_1 \pi x^2) - 0.8 \cos(f_2 \pi x^2)$ on $[-1, 1]$.

Figure	L	n	r	Target	N_{train}	Epochs	batch	fix L
5, 6	3	1024	15	$\cos(8\pi x)$	5k	10k	160	–
8	4	1024	8	(36,12)	2k	5k	100	True
9	8	1024	15	(144,48)	4k	10k	100	True
8	8	1024	20	(144,48)	4k	10k	100	False
7	4	1024	50	(36,12)	1k	1k	100	False

5.1. Log-ratio growth

We validate our theoretical predictions on frequency-dependent function approximation. We train 3-layer low-rank networks on $f(x) = \cos(8\pi x)$ with $n=1024$ neurons, $r=15$ channels, and $N=5000$ samples. We measure at $x=0$ via layer-2 low-rank channels f_k and log-ratios $R_{i,j} = \log |f_i| - \log |f_j|$. We track $\max_{i,j} R_{i,j}(x=0)$ during training (Figure 5) and the distribution over all activation pairs (Figure 6). The sustained growth of

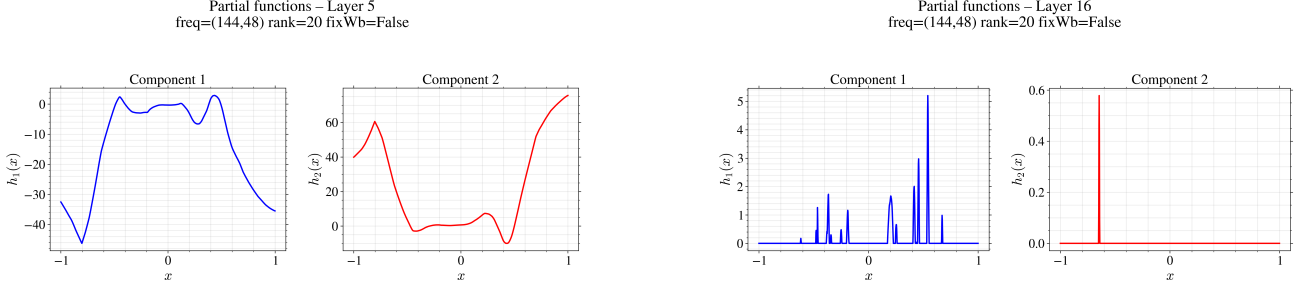


Figure 8. Asymmetry when RF is removed (a) (b) Asym. Layer 7. (c) Asym. Layer 16; $r=20$. (b)–(c) ; $f_1 = 144$, $f_2 = 48$.

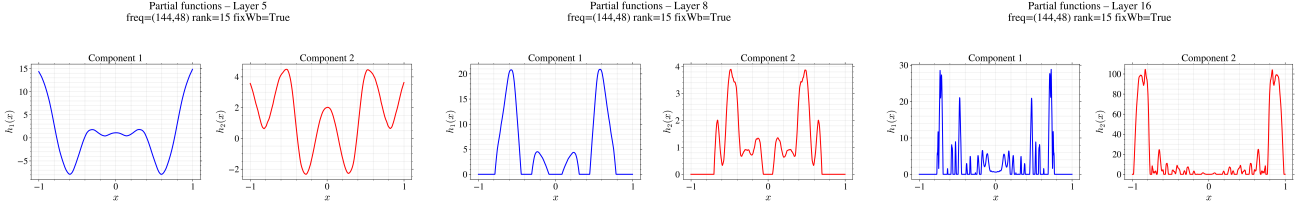


Figure 9. (symmetric) with bigger spikes. (a) $f_1^{(5)}$. (b) $\text{ReLU}(H_8)$ (c) $\text{ReLU}(H_{16})$.

$\max_{i,j} R_{i,j}$ indicates that one channel increasingly dominates at $x = 0$, consistent with Theorem 4.1. Experimental setup, ablations, and full results are in Appendix ?? and Table 2.

5.2. Low-rank learn spike features

At $x = 0$, we examine the layer-2 pre-activations H_k and the log-ratios $R_{i,j} = \log |H_i| - \log |H_j|$, using H_2 to reveal the half-space separation and discrepancy that drive spike learning. The evolution of the mean-field weight distribution over training is illustrated in Figure 10. The evolution of $\max_{i,j} R_{i,j}(x=0)$ through training, together with the full distribution of log-ratios, is shown in Figure 6. The sustained growth demonstrates increasing channel dominance at $x = 0$, in agreement with Theorem 4.1. Spike-like specialization is shown in Figure 8(a).

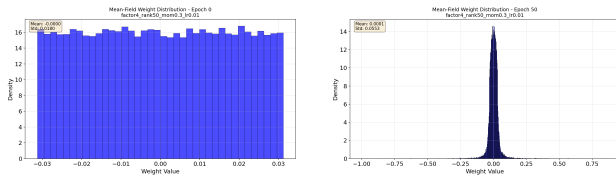


Figure 10. Mean-field weight density at epoch 0 (left) and is symmetric at 50 (right).

5.3. Channel and activations learn symmetry

We examine symmetry preservation by RF-LR on highly oscillating targets. Under batched optimization, markedly non-symmetrical features can be learned for symmetric targets. Specifically, all our targets are symmetric about $x = 0$

since both terms are even in x . When trained with SGD, low-rank networks maintain this symmetry, whereas full-rank networks show asymmetric structures. Channel feature learning in low-rank networks thus preserves geometric properties of the target, likely due to the implicit regularization of the low-rank constraint.

Training in practice. The quantitative theorem (Appendix H) provides error $O(1/\sqrt{n_{\min}})$ with Gronwall constant $e^{K_T(1+rK)}$, implying that, in worst case, training could take exponential time. However, in practical experiments, convergence is much faster due to a more favorable loss landscape shaped by hierarchical Fourier structure and plateaus at complexity transitions. Detailed, rigorous characterization of these practical speedups is left to future work (see Discussion, Section 6).

6. Conclusion

We have shown that when the mean-field dynamics converges, the limit is a global minimizer; this *persists* under low-rank constraints for mean-field networks of depth $L \geq 2$, even without convexity. Recent work has established convergence to a global minimizer for full-rank three-layer networks without convexity, but it remained open whether these guarantees hold under low-rank constraints. Our key insight is that the low-rank and random features structure are minimal to maintain the universal approximation property throughout training, which preserves the conditions needed for the limit to be a global minimizer when the dynamics converges.

Discussion: Empirically, the training loss exhibits plateaus

at each leap in complexity (e.g., when the network acquires a new frequency component), consistent with hierarchical feature learning. Rigorous characterization of this landscape structure and its implications for convergence rates is deferred to future work.

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A. Notation and Neuronal Embedding Framework

We keep the notation and neuronal embedding framework of (Nguyen & Pham, 2023). In the mean-field framework, neurons are indexed by continuous random variables rather than discrete indices. This *neuronal embedding* approach treats each neuron as a sample from a probability measure, enabling rigorous analysis in the infinite-width limit.

Neuronal indices: We use $C_1 \in \mathcal{C}_1$ and $C_2 \in \mathcal{C}_2$ to denote random variables indexing neurons in the first and second layers, respectively. These are drawn from probability measures μ_1 and μ_2 on spaces $\mathcal{C}_1$ and $\mathcal{C}_2$. In the finite-width case, $C_1(j_1)$ and $C_2(j_2)$ correspond to countable or uncountable neuron indices $j_1 \in \{1, \dots, n_1\}$ and $j_2 \in \{1, \dots, n_2\}$.

Weight functions: The weights are functions of time and neuronal indices:

- $w_1(t, C_1, k) \in \mathbb{R}$: weight for channel $k \in \{1, \dots, r\}$ of neuron C_1 in the first layer at time t .
- $w_2(t, C_2) \in \mathbb{R}$: weight for neuron C_2 in the second layer at time t .
- $W(t) = (w_1(t, \cdot, \cdot), w_2(t, \cdot))$: the full weight configuration at time t .

Feature maps and activations:

- $L^0(C_1) \in \mathbb{R}^{d_0}$: frozen random feature vector for neuron C_1 (drawn i.i.d. from a Gaussian measure); we specify the law of $L^0(C_1)$ when sampling the network.
- $\varphi_1 : \mathbb{R} \rightarrow \mathbb{R}$: activation function for the first layer (e.g., ReLU).
- $\varphi_2 : \mathbb{R} \rightarrow \mathbb{R}$: activation function for the second layer.
- $H_2(t, c_2; X, W(t)) = \sum_{k=1}^r L_{c_2, k} f_k(t; X, W(t))$: second-layer pre-activation, where $f_k(t; X, W(t)) = \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(L^0(C_1), X)]$ are the channel-wise partial functions.

Backpropagation signal (all layers): The gradient $\partial \mathcal{L} / \partial w$ at each layer is (upstream backpropagated signal) $\times$ (local derivative). Deriving $\partial_t w$ from $-\xi \mathbb{E}_Z[\dots]$ at each layer yields:

- **Top layer (layer $L - 1$):** The backpropagated signal from the loss is $D_{L-1} = d_L = \partial_{\hat{y}} \mathcal{L}(Y, \hat{y}(X; W(t)))$; it is used in the w_{L-1} -ODE.
- **Channel- k signal from layer ℓ to the layer below ($\ell = 2, \dots, L - 1$):** $B_k^{(\ell)}(t; X, W) = \mathbb{E}_{C_\ell} [L_{C_\ell, k}^{(\ell)} \varphi'_\ell(H_\ell(t, C_\ell; X, W)) w_\ell(t, C_\ell)]$, $k \in \{1, \dots, r\}$. This aggregates gradient information from layer ℓ through the mixing matrix $L^{(\ell)}$ and the activation derivative φ'_ℓ ; for ReLU, $\varphi'_\ell(u) = \mathbb{1}\{u > 0\}$ (ReLU gate). In the 3-layer case, $B_k := B_k^{(2)}$.

Loss and learning rates:

- $d_L(Z; W(t)) = \partial_{\hat{y}} \mathcal{L}(Y, \hat{y}(X; W(t)))$: loss derivative with respect to the network output, where $Z = (X, Y)$ is a data sample and $\hat{y}(X; W(t))$ is the network output.
- $\xi_1(t), \xi_2(t) \geq 0$: learning rate schedules for the first and second layers, respectively.

Kernel function:

- $K_{\mu_0}(x, X)$: kernel that measures similarity between input locations x and X , induced by the untrained first-layer feature measure μ_0 (the pushforward of the initial first-layer weights). It is defined by

$$K_{\mu_0}(x, x') \equiv \int_{\mathbb{R}^d} \varphi_1(\langle \theta, x \rangle) \varphi_1(\langle \theta, x' \rangle) \mu_0(d\theta).$$

For ReLU φ_1 and μ_0 the pushforward of i.i.d. Gaussian or uniform-on-sphere first-layer weights, K_{μ_0} coincides with the first-layer NNGP kernel (Jacot et al., 2018). This kernel appears in the evolution equation for the partial functions $f_k(t, x)$.

Expectations: $\mathbb{E}_Z[\cdot]$ denotes expectation over the data distribution, $\mathbb{E}_{C_1}[\cdot]$ over the first-layer neuron measure, and $\mathbb{E}_{C_2}[\cdot]$ over the second-layer neuron measure.

B. Assumptions

This section contains the complete statement of all assumptions used in our theoretical analysis. These assumptions are referenced in the main text with their names in italics.

Assumption B.1 (Bounded Activations and Mixing). There exists a constant $K \geq 1$ such that:

- **Activation functions:** $\|\varphi_1\|_\infty \leq K$, $\|\varphi_2\|_\infty \leq K$, and $\|\varphi'_2\|_\infty \leq K$. Moreover, φ_1 and φ_2 are K -Lipschitz continuous, and φ'_2 is non-vanishing (strictly non-zero everywhere).
- **Low-rank mixing matrix:** The mixing matrix entries $L_{c_2,k}$ are random variables (e.g., Uniform) with $\sup_{c_2,k} |L_{c_2,k}| \leq K$ almost surely, and $k \mapsto L_{c_2,k}$ is measurable for each c_2 . This implies $\|L\|_{\infty,1} \equiv \sup_{c_2} \sum_{k=1}^r |L_{c_2,k}| \leq rK$ almost surely.

Assumption B.2 (Sub-Gaussian Initialization). The initial weights satisfy sub-Gaussian moment bounds:

- For the first-layer weights: $\sup_{m \geq 1} \frac{1}{\sqrt{m}} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [|w_1^0(C_1, k)|^m]^{1/m} \leq K$ for some $K > 0$.
- For the second-layer weights: $\sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E}_{C_2} [|w_2^0(C_2)|^m]^{1/m} \leq K$.
- Equivalently, in terms of ψ_2 norms: $\|w_1(0)\|_\psi < \infty$ and $\|w_2(0)\|_\psi < \infty$, where $\|\cdot\|_\psi$ denotes the ψ_2 norm controlling moment growth.

This ensures that the initial weight distributions have controlled tail behavior, which is essential for the well-posedness argument.

Assumption B.3 (Data Distribution and Loss Regularity). • **Bounded inputs:** $|X| \leq K$ with probability 1, and the feature map satisfies $\|L^0(c_1)\| \leq K$ for all $c_1 \in \Omega_1$.

- **Loss function regularity:** The loss function $\mathcal{L}(y, \hat{y})$ is such that $\partial_2 \mathcal{L}(y, \cdot)$ is K -bounded and K -Lipschitz for all y in the support of $\mathcal{P}$, and $\partial_2 \mathcal{L}(y, u) = 0$ only when $u = 0$.
- **No convexity requirement: Crucially, we do NOT require convexity of the loss function.** This allows our results to apply to non-convex settings like frequency-dependent function approximation, where the loss function is inherently non-convex.

Assumption B.4 (Diversity of Random Features). The support of ρ^1 (the measure on Ω_1 indexing the first-layer random features) is $\mathbb{R}^d$ (or dense in $\mathbb{R}^d$). This ensures that the random features $\{\varphi_1(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\}$ have dense span in $L^2(\mathcal{P}_X)$, which is essential for the universal approximation property.

Assumption B.5 (Non-Degeneracy). To ensure that the limit point $(\bar{w}_1, \dots, \bar{w}_L)$ is non-degenerate (e.g., $\max_{1 \leq k \leq r} \mathbb{P}(\bar{w}_1(C_1, k) \neq 0) > 0$ for the first layer and $\mathbb{P}(\bar{w}_\ell(C_\ell) \neq 0) > 0$ for $\ell = 2, \dots, L$), we require one of the following:

1. The initial loss satisfies $\mathcal{L}(w_1^0, \dots, w_L^0) < \mathbb{E}_Z[\mathcal{L}(Y, \varphi_L(0))]$. Then by the gradient flow property, the limit point must have non-zero mass for each of $\bar{w}_1, \dots, \bar{w}_L$ (e.g. $\max_{1 \leq k \leq r} \mathbb{P}(\bar{w}_1(C_1, k) \neq 0) > 0$ and $\mathbb{P}(\bar{w}_\ell(C_\ell) \neq 0) > 0$ for $\ell = 2, \dots, L$), so $(\bar{w}_1, \dots, \bar{w}_L)$ is non-degenerate. This condition requires that the initial network performs better than the trivial predictor $\hat{y} = 0$, which is satisfied for most reasonable initializations (e.g., small random weights) with high probability.

Theorem B.6 (Universal approximation automatically maintained). *The learning trajectory automatically maintains the universal approximation property of the function class represented by the first layer's neurons throughout training. Specifically, if $\text{supp}(L^0(C_1)) = \mathbb{R}^d$ and φ_1 is ReLU (or any non-polynomial activation), then the function class $\{\varphi_1(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\}$ has dense span in $L^2(\mathcal{P}_X)$ throughout training.*

This follows from the fact that since $L^0(C_1)$ are frozen random features with full support, and ReLU is non-polynomial, the dense span property is automatically maintained: $\text{ReLU}(af + b)$ for random a, b always forms a dense span. This is the key property that, combined with low-rank structure, enables convergence to a global minimizer even without convexity.

Assumption B.7 (Convergence to Limit Point). There exist functions $\bar{w}_1 : \Omega_1 \times \{1, \dots, r\} \rightarrow \mathbb{R}$ and $\bar{w}_2 : \Omega_2 \rightarrow \mathbb{R}$ such that as $t \rightarrow \infty$, there exists a coupling π_t of $\rho^1 \times \rho^2$ and itself such that:

$$\int (1 + |\bar{w}_2(c_2)|) |\bar{w}_2(c_2)| \max_{1 \leq k \leq r} |\bar{w}_1(c_1, k)| |w_1^*(t, c'_1, k) - \bar{w}_1(c_1, k)| d\pi_t(c_1, c_2, c'_1, c'_2) \rightarrow 0, \quad (5)$$

$$\int (1 + |\bar{w}_2(c_2)|) |\bar{w}_2(c_2)| |w_2^*(t, c'_2) - \bar{w}_2(c_2)| d\pi_t(c_1, c_2, c'_1, c'_2) \rightarrow 0, \quad (6)$$

where $W^*(t) = (w_1^*(t, \cdot, \cdot), w_2^*(t, \cdot))$ is the solution to the mean-field ODEs (2). This assumption ensures that the training dynamics converge to a well-defined limit point in a Wasserstein-like sense.

Remark B.8 (On the assumptions). Universal approximation is automatically maintained (Theorem B.6), which is the crucial difference from previous work. Rather than relying on convexity of the loss, we prove that the function class maintains its approximation power throughout training. The frozen random features in the first layer with full support automatically ensure dense span: since $\text{ReLU}(af + b)$ for random a, b always forms a dense span, this property is automatically maintained rather than assumed, providing a rich fixed basis independent of the low-rank structure in subsequent layers.

The *Convergence to Limit Point* assumption is typically verified by showing that the loss decreases along trajectories (gradient flow property), establishing compactness of the trajectory set, and using stability arguments such as LaSalle's invariance principle. The low-rank structure enters through the $\max_k$ over r channels in (5), but the overall structure remains similar to the full-rank case.

C. Technical Lemmas and Proofs for Well-Posedness

The lemmas below and the techniques underlying the assumptions in the previous section are from the rigorous mean-field framework of (Nguyen & Pham, 2023). We adapt them to the low-rank case by accounting for the r channels and the mixing matrix L (e.g., through $\|L\|_{\infty,1} \leq rK$).

C.1. Key New Lemma: Bi-Lipschitz Property

Before proving well-posedness, we establish a key technical lemma that will be used throughout the proof.

Lemma C.1 (Bi-Lipschitz property of H_ℓ , $\ell = 2, \dots, L-1$). *For any $W' = (w'_1, \dots, w'_{L-1})$ and $W'' = (w''_1, \dots, w''_{L-1})$ (in the 3-layer case $W = (w_1, w_2)$) satisfying the regularity assumptions, we have the following for each $\ell = 2, \dots, L-1$.*

Layer $\ell = 2$.

$$|H_2(t, c_2; X, W') - H_2(t, c_2; X, W'')| \leq K \|L^{(2)}\|_{\infty,1} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [|w'_1(t, C_1, k) - w''_1(t, C_1, k)|],$$

where $\|L^{(\ell)}\|_{\infty,1} \equiv \sup_{c_\ell} \sum_{k=1}^r |L_{c_\ell, k}^{(\ell)}| \leq rK$ under the entrywise bound ($L^{(2)} = L$). If φ_2 is ReLU or Lipschitz, then

$$|\varphi_2(H_2(t, c_2; X, W')) - \varphi_2(H_2(t, c_2; X, W''))| \leq K \|L^{(2)}\|_{\infty,1} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [|w'_1(t, C_1, k) - w''_1(t, C_1, k)|].$$

Layers $\ell = 3, \dots, L-1$.

$$|H_\ell(t, c_\ell; X, W') - H_\ell(t, c_\ell; X, W'')| \leq K \|L^{(\ell)}\|_{\infty,1} \left(\mathbb{E}_{C_{\ell-1}} [|w'_{\ell-1}(t, C_{\ell-1}) - w''_{\ell-1}(t, C_{\ell-1})|] + \llbracket w''_{\ell-1} \rrbracket_t \cdot \mathcal{B}_{\ell-1}(W', W'') \right),$$

where $\mathcal{B}_{\ell-1}(W', W'')$ is the RHS of the bi-Lipschitz inequality for $H_{\ell-1}$ above. If φ_ℓ is ReLU or Lipschitz, the same bound holds for $|\varphi_\ell(H_\ell(t, c_\ell; X, W')) - \varphi_\ell(H_\ell(t, c_\ell; X, W''))|$.

Proof. Layer $\ell = 2$. By definition $H_2(t, c_2; X, W) = \sum_{k=1}^r L_{c_2, k} f_k(t; X, W)$ with $f_k(t; X, W) = \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), X \rangle)]$. Then

$$\begin{aligned} H_2(t, c_2; X, W') - H_2(t, c_2; X, W'') &= \sum_{k=1}^r L_{c_2, k} (f_k(t; X, W') - f_k(t; X, W'')) \\ &= \sum_{k=1}^r L_{c_2, k} \mathbb{E}_{C_1} [(w'_1(t, C_1, k) - w''_1(t, C_1, k)) \varphi_1(\langle L^0(C_1), X \rangle)]. \end{aligned}$$

Taking absolute values and using boundedness of φ_1 :

$$\begin{aligned} |H_2(t, c_2; X, W') - H_2(t, c_2; X, W'')| &\leq \sum_{k=1}^r |L_{c_2, k}| \mathbb{E}_{C_1} [|w'_1(t, C_1, k) - w''_1(t, C_1, k)|] |\varphi_1(\langle L^0(C_1), X \rangle)| \\ &\leq K \sum_{k=1}^r |L_{c_2, k}| \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [|w'_1(t, C_1, k) - w''_1(t, C_1, k)|] \\ &\leq K \|L^{(2)}\|_{\infty, 1} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [|w'_1(t, C_1, k) - w''_1(t, C_1, k)|]. \end{aligned}$$

The $\varphi_2(H_2)$ part follows from the Lipschitz property of φ_2 .

Layers $\ell \geq 3$. We have $H_\ell(t, c_\ell; X, W) = \sum_{k=1}^r L_{c_\ell, k}^{(\ell)} f_k^{(\ell)}(t; X, W)$ with $f_k^{(\ell)}(t; X, W) = \mathbb{E}_{C_{\ell-1}} [w_{\ell-1}(t, C_{\ell-1}, k) \varphi_{\ell-1}(H_{\ell-1}(t, C_{\ell-1}; X, W))]$ (and $w_{\ell-1}(t, C_{\ell-1}, k) \equiv w_{\ell-1}(t, C_{\ell-1})$ when the layer has no channel index). Then

$$f_k^{(\ell)}(W') - f_k^{(\ell)}(W'') = \mathbb{E}_{C_{\ell-1}} [(w'_{\ell-1} - w''_{\ell-1}) \varphi_{\ell-1}(H_{\ell-1}(W')) + w''_{\ell-1} (\varphi_{\ell-1}(H_{\ell-1}(W')) - \varphi_{\ell-1}(H_{\ell-1}(W'')))].$$

Using $|\varphi_{\ell-1}| \leq K$, $|\varphi_{\ell-1}(a) - \varphi_{\ell-1}(b)| \leq K|a - b|$, and the inductive bound for $|H_{\ell-1}(W') - H_{\ell-1}(W'')|$:

$$\begin{aligned} |f_k^{(\ell)}(W') - f_k^{(\ell)}(W'')| &\leq K \mathbb{E}_{C_{\ell-1}} [|w'_{\ell-1} - w''_{\ell-1}|] + K \mathbb{E}_{C_{\ell-1}} [|w''_{\ell-1}| |H_{\ell-1}(W') - H_{\ell-1}(W'')|] \\ &\leq K \mathbb{E}_{C_{\ell-1}} [|w'_{\ell-1} - w''_{\ell-1}|] + K \|w''_{\ell-1}\|_t \mathcal{B}_{\ell-1}(W', W''). \end{aligned}$$

Thus $|H_\ell(W') - H_\ell(W'')| \leq \|L^{(\ell)}\|_{\infty, 1} \max_k |f_k^{(\ell)}(W') - f_k^{(\ell)}(W'')|$ yields the claim. The $\varphi_\ell(H_\ell)$ part follows from the Lipschitz property of φ_ℓ . $\square$

From here, the proof follows as usual. For each $\ell = 2, \dots, L-1$, $H_\ell(t, c_\ell; X, W)$ bi-Lipschitz in W is sufficient and equivalent to φ_ℓ Lipschitz; that is the only thing that matters. The remaining is simple: after defining norms, we update $K_0(t)$ with a factor $(1 + rK)^{1/2}$; the solution operator F , a priori bounds, and the contraction argument then proceed as in the full-rank framework (Nguyen & Pham, 2023).

C.2. Norms and Spaces

We equip the mean-field parameters with several norms. For the low-rank case: $\|w_1\|_t = \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [\sup_{s \leq t} |w_1(s, C_1, k)|^{50}]^{1/50}$, $\|w_2\|_t = \mathbb{E}_{C_2} [\sup_{s \leq t} |w_2(s, C_2)|^{50}]^{1/50}$, $\|W\|_t = \max(\|w_1\|_t, \|w_2\|_t)$. L^2 -type: $\|w_1\|_t = \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [\sup_{s \leq t} |w_1(s, C_1, k)|^2]^{1/2}$, $\|w_2\|_t = \mathbb{E}_{C_2} [\sup_{s \leq t} |w_2(s, C_2)|^2]^{1/2}$, $\|W\|_t = \max(\|w_1\|_t, \|w_2\|_t)$. ψ_2 -type: $\|w_1\|_{\psi, t} = \sqrt{50} \sup_{m \geq 1} \frac{1}{\sqrt{m}} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [\sup_{s \leq t} |w_1(s, C_1, k)|^m]^{1/m}$, $\|w_2\|_{\psi, t} = \sqrt{50} \sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E}_{C_2} [\sup_{s \leq t} |w_2(s, C_2)|^m]^{1/m}$, $\|W\|_{\psi, t} = \max(\|w_1\|_{\psi, t}, \|w_2\|_{\psi, t})$. The factor $\sqrt{50}$ ensures $\|W\|_{\psi, t} \geq \|W\|_t$ and $\|W\|_{\psi, t} \geq \|W\|_t$. Random variables: $\max_t^w(W) = \max_{1 \leq k \leq r} \sup_{s \leq t} |w_1(s, C_1, k)|$, $\max_t^{w_2}(W) = \sup_{s \leq t} |w_2(s, C_2)|$. Distance for W, W' :

$$\|W - W'\|_t = \max(\|w_1 - w'_1\|_t, \|w_2 - w'_2\|_t), \quad (7)$$

with $\|w_1 - w'_1\|_t = \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [\sup_{s \leq t} |w_1(s, C_1, k) - w'_1(s, C_1, k)|^2]^{1/2}$ and $\|w_2 - w'_2\|_t = \mathbb{E}_{C_2} [\sup_{s \leq t} |w_2(s, C_2) - w'_2(s, C_2)|^2]^{1/2}$.

C.3. Solution Operator and Fixed Point Formulation

Denote by $\mathfrak{W}_T$ the space of mean-field parameters W with $\|W\|_T < \infty$. Given $T \geq 0$ and $W(0)$, we define F mapping $W' \in \mathfrak{W}_T$ to $F(W')(t) = \{F_1(W')(t, \cdot, \cdot), F_2(W')(t, \cdot, \cdot)\}$, where $F_1(W')(t, c_1, k) = w_1(0, c_1, k) - \int_0^t \xi_1(s) \mathbb{E}_Z [d_L(Z; W'(s)) \varphi_1(\langle L^0(c_1), X \rangle) B_k(s; X, W'(s))] ds$ and $F_2(W')(t, c_2) = w_2(0, c_2) - \int_0^t \xi_2(s) \mathbb{E}_Z [d_L(Z; W'(s)) \varphi_2(H_2(s, c_2; X, W'(s)))] ds$, with $B_k(s; X, W') = \mathbb{E}_{C_2} [L_{C_2, k} \varphi_2'(H_2(s, C_2; X, W')) w'_2(s, C_2)]$. At initialization $F(W')(0, \cdot, \cdot) = W(0)$; the time integrals use W' . A solution on $[0, T]$ is $W \in \mathfrak{W}_T$ with $F(W) = W$. We say W is a solution on $[0, \infty)$ if its restriction to $[0, T]$ is a solution for all $T > 0$.

C.4. A Priori Bounds

Lemma C.2 (Weight bounds). *Under Assumptions B.1 and B.3, given an initialization $W(0)$, a solution W to the mean-field ODEs, if it exists, must satisfy that for any $t \in [0, \infty)$:*

$$\|W\|_t \leq K_0(t),$$

where $K_0(t)$ is a non-decreasing function of the form

$$K_0(t) = (1 + rK)^{1/2} K^\kappa (1 + t^\kappa) (1 + \|W\|_0^\kappa),$$

for some constant $\kappa > 0$ depending on K and r .

A similar result holds for the ψ_2 norm. Under the same assumptions, for any $t \in [0, \infty)$, there exists $K_0(t) \geq 1$ of the form

$$K_0(t) = (1 + rK)^{1/2} K^\kappa (1 + t^\kappa) (1 + \|W\|_{\psi,0}^\kappa),$$

such that a solution W , if it exists, must satisfy $\|W\|_t \leq \|W\|_{\psi,t} \leq K_0(t)$ for any $t \in [0, \infty)$. Furthermore, by assuming $\|W\|_{\psi,0} < \infty$, for any $B \geq 0$:

$$\mathbb{P}(\max(\max_t^w(W), \max_t^{w_2}(W)) \geq K_0(t)B) \leq Ce^{-K_1 B^2},$$

for some universal constants $C, K_1 > 0$.

Proof. The proof follows from Grönwall-type arguments applied to the ODEs (2). The key steps are:

1. Use the boundedness and Lipschitz properties of $\varphi_1, \varphi_2, \varphi'_2$, and $\partial_2 \mathcal{L}$.
2. Control H_2 using the low-rank structure: $|H_2| \leq \|L\|_{\infty,1} \max_k |f_k| \leq rK \max_k \mathbb{E}_{C_1}[|w_1(C_1, k)|]$.
3. Control B_k using the structure: $|B_k| \leq K \mathbb{E}_{C_2}[|L_{C_2,k}| |w_2|] \leq rK^2 \mathbb{E}_{C_2}[|w_2|]$.
4. Apply Minkowski's inequality and Grönwall's lemma to obtain polynomial growth in t .
5. The sub-Gaussian tail bound follows from the ψ_2 control and a union bound.

The constant κ depends on K and r through the factor $\|L\|_{\infty,1} \leq rK$, but remains polynomial rather than exponential in r . $\square$

These a priori bounds lead us to consider the following spaces, given an initialization $W(0)$ and an arbitrary terminal time $T > 0$:

- The space $\mathcal{W}_T$ of mean-field parameters $W' = \{W'(t)\}_{t \leq T}$ such that $\|W'\|_T \leq K_0(T)$.
- The space $\mathcal{W}_T^0 \subset \mathcal{W}_T$ of mean-field parameters $W' \in \mathcal{W}_T$ such that:

$$\|W'\|_{\psi,T} \leq K_0(T),$$

$$\mathbb{P}(\max(\max_T^w(W'), \max_T^{w_2}(W')) \geq K_0(T)B) \leq Ce^{-K_1 B^2} \quad \forall B \geq 0,$$

and $W'(0) = W(0)$ (so all elements in $\mathcal{W}_T^0$ share the same initialization).

It is easy to see that $\mathcal{W}_T^0 \subset \mathcal{W}_T$ since $\|W'\|_T \leq \|W'\|_{\psi,T}$.

We equip these spaces with the metric $(W', W'') \mapsto \|W' - W''\|_T$. By Lemma C.2, any solution W to the mean-field ODEs, if it exists, must belong to $\mathcal{W}_T^0$.

Lemma C.3 (Solution operator maps bounded sets to bounded sets). *Under Assumptions B.1 and B.3, for any $W' \in \mathcal{W}_T^0$, we have $F(W') \in \mathcal{W}_T^0$.*

Proof. The proof follows the same argument as Lemma C.2, using the integral form defining F and the bounded/Lipschitz properties of the drifts. The key is that applying bounded/Lipschitz drifts to W' preserves the moment and tail bounds with constants controlled by $K_0(T)$. $\square$

C.5. Difference Estimate

Lemma C.4 (Solution operator is contractive). *For a given $B \geq 0$, consider two collections of mean-field parameters $W', W'' \in \mathcal{W}_T$ such that:*

$$\begin{aligned} \mathbb{P}(\max(\max_T^w(W'), \max_T^{w_2}(W')) \geq K_0(T)B) &\leq Ce^{-K_1B^2}, \\ \mathbb{P}(\max(\max_T^w(W''), \max_T^{w_2}(W'')) \geq K_0(T)B) &\leq Ce^{-K_1B^2}. \end{aligned}$$

Under Assumptions B.1–B.3, for any $t \leq T$:

$$\|F(W') - F(W'')\|_t \leq (KK_0(T))^4 \int_0^t \left((1+B)\|W' - W''\|_s + \sqrt{2}e^{-K_1B^2/2} \right) ds,$$

where the constant depends on K and r through $\|L\|_{\infty,1} \leq rK$.

Proof. The proof uses a good/bad event decomposition:

1. On the good event $\{\max(\max_T^w, \max_T^{w_2}) \leq K_0(T)B\}$, the drifts are Lipschitz in W with constant $(1+B)K_0(T)$.
2. The difference $|H_2(W') - H_2(W'')|$ is controlled using Lemma C.1:

$$|H_2(W') - H_2(W'')| \leq K\|L\|_{\infty,1} \max_k \mathbb{E}_{C_1} [|w'_1(C_1, k) - w''_1(C_1, k)|] \leq rK^2 \max_k \|w'_1 - w''_1\|_t.$$

3. The difference $|B_k(W') - B_k(W'')|$ is controlled similarly, giving a factor $(1+B)$.
4. The bad event contributes an exponentially small remainder $e^{-K_1B^2/2}$.
5. Integrating in time and using Minkowski's inequality yields the result.

□

C.6. Complete Proof of Theorem 3.1

Proof of Theorem 3.1. We perform a Picard-type iteration argument. Consider an arbitrary finite $T \geq 0$ and $W', W'' \in \mathcal{W}_T^0$. From Lemma C.4:

$$\begin{aligned} \|F(W') - F(W'')\|_t &\leq (KK_0(T))^4 \left((1+B) \int_0^t \|W' - W''\|_s ds + T\sqrt{2}e^{-K_1B^2/2} \right) \\ &\equiv k_1(1+B) \int_0^t \|W' - W''\|_s ds + k_2e^{-k_3B^2}, \end{aligned}$$

for any $B > 0$, where $k_1 = (KK_0(T))^4$, $k_2 = (KK_0(T))^4 T\sqrt{2}$, and $k_3 = K_1/2$.

By Lemma C.3, F maps $\mathcal{W}_T^0$ to $\mathcal{W}_T^0$. We can iterate this inequality to obtain:

$$\begin{aligned} \|F^{(m)}(W') - F^{(m)}(W'')\|_T &\leq k_1(1+B) \int_0^T \|F^{(m-1)}(W') - F^{(m-1)}(W'')\|_{T_2} dT_2 + k_2e^{-k_3B^2} \\ &\leq k_1^2(1+B)^2 \int_0^T \int_0^{T_2} \|F^{(m-2)}(W') - F^{(m-2)}(W'')\|_{T_3} \mathbb{I}(T_2 \leq T) dT_3 dT_2 \\ &\quad + k_2 \sum_{\ell=1}^2 \frac{(Tk_1(1+B))^{\ell-1}}{\ell!} e^{-k_3B^2} \\ &\quad \dots \\ &\leq \frac{1}{m!} T^m k_1^m (1+B)^m \|W' - W''\|_T + k_2 e^{Tk_1(1+B) - k_3B^2} \\ &\leq \frac{1}{m!} T^m k_1^m (1+\sqrt{m})^m \|W' - W''\|_T + k_2 e^{Tk_1(1+\sqrt{m}) - k_3m}, \end{aligned}$$

where we choose $B = \sqrt{m}$ in the last display. Note that since $\|W\|_0 < \infty$, $K_0(T)$ and hence k_1, k_2 are finite for finite T .

By substituting $W'' = F(W')$, we obtain:

$$\sum_{m=1}^{\infty} \|F^{(m+1)}(W') - F^{(m)}(W')\|_T = \sum_{m=1}^{\infty} \|F^{(m)}(W'') - F^{(m)}(W')\|_T < \infty.$$

Hence as $m \rightarrow \infty$, $F^{(m)}(W')$ converges in $\|\cdot\|_T$ to a limit $W \in \mathfrak{W}_T$, which is a fixed point of F . By Lemma C.2, W belongs to $\mathcal{W}_T^0$.

The uniqueness of the fixed point comes from the above estimate, since if W' and W'' are fixed points of F , then they are both in $\mathcal{W}_T^0$, and:

$$\|W' - W''\|_T = \|F^{(m)}(W') - F^{(m)}(W'')\|_T \leq \frac{1}{m!} T^m k_1^m (1 + \sqrt{m})^m \|W' - W''\|_T + k_2 e^{T k_1 (1 + \sqrt{m}) - k_3 m},$$

and one can take m arbitrarily large. This proves that the solution exists and is unique on $t \in [0, T]$. Since T is arbitrary, we have existence and uniqueness of the solution to the mean-field ODEs on the time interval $[0, \infty)$. $\square$

D. Lemma and proofs of convergence

D.1. Channel Mixing and Low-Rank Structure

The low-rank channel mixing structure enters through:

$$H_2(t, c_2; X, W) = \sum_{k=1}^r L_{c_2, k} f_k(t; X, W),$$

where $f_k(t; X, W) = \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), X \rangle)]$ are the r partial functions.

Under the entrywise bound $\sup_{c_2, k} |L_{c_2, k}| \leq K$, we have:

$$\|L\|_{\infty, 1} \leq rK, \quad \|L\|_{\infty, 2} \leq \sqrt{r} K.$$

Lemma D.1 (Sub-Gaussian bounds for low-rank sums). *Let $(U_k)_{k=1}^r$ be real random variables on a common probability space and $(a_k)_{k=1}^r \in \mathbb{R}^r$ deterministic. Then for every $m \geq 1$,*

$$\mathbb{E} \left[\left| \sum_{k=1}^r a_k U_k \right|^m \right]^{1/m} \leq \sum_{k=1}^r |a_k| \mathbb{E}[|U_k|^m]^{1/m} \leq \left(\sum_{k=1}^r |a_k| \right) \max_{1 \leq k \leq r} \mathbb{E}[|U_k|^m]^{1/m}.$$

Consequently, any “ ψ_2 -type” seminorm defined by $\sup_{m \geq 1} m^{-1/2} \mathbb{E}[|\cdot|^m]^{1/m}$ satisfies

$$\sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E} \left[\left| \sum_{k=1}^r a_k U_k \right|^m \right]^{1/m} \leq \left(\sum_{k=1}^r |a_k| \right) \max_{1 \leq k \leq r} \sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E}[|U_k|^m]^{1/m}.$$

If one prefers an ℓ_2 version, then also $\sum_{k=1}^r |a_k| \leq \sqrt{r} (\sum_k a_k^2)^{1/2}$ yields a $\sqrt{r}$ factor.

Lemma D.2 (Forward propagation bounds scale with mixing matrix norm). *Assume φ_1 is bounded by K . Then for every $t \geq 0$ and every $c_2 \in \Omega_2$,*

$$\sup_{s \leq t} |H_2(s, c_2; X, W)| \leq \left(\sum_{k=1}^r |L_{c_2, k}| \right) \max_{1 \leq k \leq r} \sup_{s \leq t} |f_k(s; X, W)| \leq \|L\|_{\infty, 1} \max_{1 \leq k \leq r} \sup_{s \leq t} |f_k(s; X, W)|.$$

Moreover,

$$\sup_{s \leq t} |f_k(s; X, W)| \leq \mathbb{E}_{C_1} \left[\sup_{s \leq t} |w_1(s, C_1, k)| |\varphi_1(\langle L^0(C_1), X \rangle)| \right] \leq K \mathbb{E}_{C_1} \left[\sup_{s \leq t} |w_1(s, C_1, k)| \right].$$

Combining these inequalities and taking moments yields, for every $m \geq 1$,

$$\mathbb{E}_X \left[\sup_{s \leq t} \sup_{c_2} |H_2(s, c_2; X, W)|^m \right]^{1/m} \leq K \|L\|_{\infty, 1} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} \left[\sup_{s \leq t} |w_1(s, C_1, k)|^m \right]^{1/m}.$$

In particular, in the ψ_2 -type calibration,

$$\sqrt{50} \sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E}_X \left[\sup_{s \leq t} \sup_{c_2} |H_2(s, c_2; X, W)|^m \right]^{1/m} \leq K \|L\|_{\infty, 1} \llbracket w_1 \rrbracket_{\psi, t}.$$

D.2. Proof of Theorem 3.2 (global minimizer at limit, any depth)

This subsection adapts the core argument of (Nguyen & Pham, 2023), Sec. 6.3 (Proof of Theorem 34), to our low-rank setting for any depth $L \geq 2$. We omit the homotopy argument (their Lemma 37): the first-layer feature map $L^0(C_1)$ is frozen, so $\text{supp}(L^0(C_1)) = \mathbb{R}^d$ at init and at all t , and dense span follows without a homotopy. The main-text

High-level proof idea and (Nguyen & Pham, 2023) Sec. 6.2.1 summarize the idea; we give the formal steps.

Dense span without homotopy. $L^0(C_1)$ is not trained. By Assumption B.4 and Theorem B.6, $\{\varphi_1(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\}$ has dense span in $L^2(\mathcal{P}_X)$ at all $t \geq 0$ and at the limit.

Zero derivative at the limit. Let $\bar{W} = (\bar{w}_1, \dots, \bar{w}_{L-1})$ be a limit point under Assumption B.7 (and its natural extension to L layers with couplings for all w_ℓ). At the limit, $\partial_t w_\ell = 0$ for all ℓ . From (2), for the top layer $\ell = L - 1$:

$$\mathbb{E}_Z [d_L(Z; \bar{W}) \varphi_{L-1}(H_{L-1}(c_{L-1}; X, \bar{W}))] = 0, \quad \forall c_{L-1}.$$

By backpropagation, the first-layer ODE yields

$$\mathbb{E}_Z [d_L(Z; \bar{W}) \varphi_1(\langle L^0(c_1), X \rangle) B_k^{(2)}(X; \bar{W})] = 0, \quad c_1 \in \text{supp}(\rho^1), \quad k \in \{1, \dots, r\},$$

where $B_k^{(2)}(X; \bar{W}) = \mathbb{E}_{C_2} [L_{C_2, k} \varphi'_2(H_2(C_2; X, \bar{W})) \bar{w}_2(C_2)]$ (and for $L > 3$, $B_k^{(\ell)}$ involves $H_\ell, \bar{w}_\ell, L^{(\ell)}$ in the same way). Since $B_k^{(2)}$ depends only on X and $\bar{W}$, we have

$$\mathbb{E}_X [\mathbb{E}_{Y|X} [d_L(Z; \bar{W}) | X] \varphi_1(\langle L^0(c_1), X \rangle) B_k^{(2)}(X; \bar{W})] = 0 \quad \forall c_1, k.$$

The function $x \mapsto \mathbb{E}_{Y|X} [d_L(Z; \bar{W}) | X = x] B_k^{(2)}(x; \bar{W})$ thus has zero inner product in $L^2(\mathcal{P}_X)$ with $\varphi_1(\langle L^0(c_1), \cdot \rangle)$ for all c_1 . By dense span, $\mathbb{E}_{Y|X} [d_L(Z; \bar{W}) | X = x] B_k^{(2)}(x; \bar{W}) = 0$ for $\mathcal{P}_X$ -a.e. x .

From integrated identity to $\partial_{\hat{y}} \mathcal{L}$ a.e. Under Assumption B.5, $\max_{1 \leq k \leq r} \mathbb{P}(\bar{w}_1(C_1, k) \neq 0) > 0$ and $\mathbb{P}(\bar{w}_\ell(C_\ell) \neq 0) > 0$ for $\ell = 2, \dots, L - 1$; φ'_ℓ is non-vanishing (Assumption B.1). Hence for at least one k , $B_k^{(2)}(x; \bar{W})$ is non-zero on a set of positive $\mathcal{P}_X$ -measure. On that set, $\mathbb{E}_{Y|X} [d_L(Z; \bar{W}) | X = x] = 0$, and by the same logic applied across x , $\mathbb{E}_Z [\partial_{\hat{y}} \mathcal{L}(Y, \hat{y}(X; \bar{W})) | X = x] = 0$ for $\mathcal{P}_X$ -a.e. x .

Case 1 (convex) and Case 2 ($y = y(x)$). If $\mathcal{L}$ is convex in the second variable, then for any measurable $\tilde{y}(x)$, $\mathcal{L}(y, \tilde{y}(x)) - \mathcal{L}(y, \hat{y}(x; \bar{W})) \geq \partial_2 \mathcal{L}(y, \hat{y}(x; \bar{W}))(\tilde{y}(x) - \hat{y}(x; \bar{W}))$. Taking expectation and using $\mathbb{E}_Z [\partial_2 \mathcal{L}(Y, \hat{y}(X; \bar{W})) | X] = 0$ a.e., we get $\mathcal{L}(\bar{W}) \leq \mathbb{E}_Z [\mathcal{L}(Y, \tilde{y}(X))]$, so $\bar{W}$ is a global minimizer. If $y = y(x)$ (a.e.), then $\partial_2 \mathcal{L}(y, \hat{y}(x; \bar{W})) = 0$ a.e. implies $\mathcal{L}(y, \hat{y}(x; \bar{W})) = 0$ $\mathcal{P}_X$ -a.e., hence $\mathcal{L}(\bar{W}) = 0$.

$\mathcal{L}(W(t)) \rightarrow \mathcal{L}(\bar{W})$. By Assumption B.7, the couplings π_t and the Wasserstein-like integrals (5)–(6) (and their L -layer analogues) tend to 0. The output difference $|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|$ is bounded by a K -multiple of those integrals (via the low-rank structure: $H_2 = \sum_k L_{c_2, k} f_k$, $B_k^{(\ell)}$, and the regularity of φ_ℓ , $\partial_2 \mathcal{L}$). Thus $\mathcal{L}(W(t)) - \mathcal{L}(\bar{W}) = \mathbb{E}_Z [\mathcal{L}(Y, \hat{y}(X; W(t))) - \mathcal{L}(Y, \hat{y}(X; \bar{W}))]$ is bounded by $K \mathbb{E}_Z [|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \rightarrow 0$ as $t \rightarrow \infty$.

E. Finite-support reduction and scalar ODEs

This appendix collects the general setup that justifies the *two-sided step* toy model and the reduction of spike learning to scalar ODEs. The w_1 evolution is linear in feature space once residuals and backprop signals are fixed; for finitely-supported data one can solve f_k exactly as a superposition of kernel bumps.

Origin of K_{μ_0} . The kernel K_{μ_0} arises from the mean-field $\partial_t f_k$ evolution. One has $f_k(t, x) = \int \bar{w}_{1,k}(t, \theta) \varphi_1(\langle \theta, x \rangle) \mu_0(d\theta)$ with $\mu_0 = (w_0)_{\#} \rho^1$ the pushforward of the untrained first-layer feature map. The w_1 -ODE yields $\partial_t \bar{w}_{1,k}(t, \theta) = -\xi_1(t) \mathbb{E}_{Z=(X,Y)}[d_L \varphi_1(\langle \theta, X \rangle) B_k(t; X)]$; differentiating under the integral gives $\partial_t f_k(t, x) = -\xi_1(t) \mathbb{E}_Z[d_L B_k(t; X) K_{\mu_0}(x, X)]$ with

$$K_{\mu_0}(x, x') \equiv \int_{\mathbb{R}^d} \varphi_1(\langle \theta, x \rangle) \varphi_1(\langle \theta, x' \rangle) \mu_0(d\theta).$$

For ReLU φ_1 and μ_0 the pushforward of i.i.d. Gaussian or uniform-on-sphere first-layer weights, K_{μ_0} coincides with the first-layer NNGP kernel (Jacot et al., 2018). We do not assume K_{μ_0} is diagonal-dominant on a full Gram matrix; for the two-point support $\{x_0, x_1\}$ we only use that $|K_{\mu_0}(x_0, x_1)| = O(e^{-c\delta})$ (Laplace-like decay for separated points), which holds for such NNGP kernels in 1D or under suitable geometry.

Finite-support data. Assume the input marginal is supported on $x^{(1)}, \dots, x^{(m)}$ with $\mathbb{P}(X = x^{(p)}) = \pi_p$ and targets $y^{(p)}$. Set $d_p(t) = d_L((x^{(p)}, y^{(p)}); W(t))$ and $B_{k,p}(t) = B_k(t; x^{(p)})$. The mean-field w_1 -PDE integrated in time yields

$$\Gamma_{k,p}(t) = \int_0^t \xi_1(s) \pi_p d_p(s) B_{k,p}(s) ds,$$

and

$$f_k(t, x) = f_k(0, x) - \sum_{p=1}^m \Gamma_{k,p}(t) K_{\mu_0}(x, x^{(p)}), \quad k = 1, \dots, r.$$

The spike shape is determined by K_{μ_0} ; all learning dynamics reduce to the scalar coefficients $\Gamma_{k,p}(t)$.

One-spike reduction. If $m = 1$ with $x^{(1)} = x_*$, then $f_k(t, x) = f_k(0, x) - \Gamma_{k,1}(t) K_{\mu_0}(x, x_*)$. For $f_k(0, \cdot) \equiv 0$, $f_k(t, \cdot)$ is exactly a kernel bump at x_* and spike learning reduces to solving $\Gamma_{k,1}(t)$.

Two-sided step ($m = 2$). Take $x^{(1)} = x_0 = -\delta$, $x^{(2)} = x_1 = +\delta$ ($\delta > 0$), $y^{(1)} = +A$, $y^{(2)} = -A$. Then

$$f_k(t, x) = f_k(0, x) - \Gamma_{k,1}(t) K_{\mu_0}(x, x_0) - \Gamma_{k,2}(t) K_{\mu_0}(x, x_1).$$

The kernel K_{μ_0} (NNGP, (Jacot et al., 2018)) is Laplace-like: $K_{\mu_0}(x_p, x_p) = K_0 > 0$ for $p \in \{0, 1\}$, and $|K_{\mu_0}(x_0, x_1)| = |K_{\mu_0}(-\delta, +\delta)| \leq C e^{-c\delta}$ for some $C, c > 0$. The full evolution is

$$\partial_t f_k(t, x_p) = -\xi_1(t) K_{\mu_0}(x_p, x_p) d_p(t) B_{k,p+1}(t) + E_p(t),$$

with $|E_p(t)| \leq C' e^{-c\delta}$. This is the setting of Theorem 4.1.

F. Proof of Theorem 4.1 (two-sided step, x_0)

We prove the theorem at x_0 ; the argument at x_1 is symmetrical.

At x_0 . By the log-ratio derivative identity at x_0 ,

$$\partial_t R_{12}(t, x_0) = \frac{\text{sign}(f_1) \partial_t f_1}{|f_1|} - \frac{\text{sign}(f_2) \partial_t f_2}{|f_2|}.$$

Insert the full evolution (4): $\partial_t f_k(t, x_0) = -\xi_1 K_{\mu_0}(x_0, x_0) d_0 B_{k,1} + E_0(t)$ with $|E_0(t)| \leq C' e^{-c\delta}$. Then

$$\partial_t R_{12}(t, x_0) = \xi_1(t) K_{\mu_0}(x_0, x_0) (-d_0(t)) \left(\frac{\text{sign}(f_1) B_{1,1}}{|f_1|} - \frac{\text{sign}(f_2) B_{2,1}}{|f_2|} \right) + \varepsilon_0(t), \quad |\varepsilon_0(t)| \leq C'' e^{-c\delta},$$

where $\varepsilon_0(t) = E_0(t) \left(\frac{\text{sign}(f_1)}{|f_1|} - \frac{\text{sign}(f_2)}{|f_2|} \right)$ inherits the bound from E_0 . Use $-d_0 \geq 0$. The sign condition gives $\text{sign}(f_1)B_{1,1} = |B_{1,1}|$; $-\text{sign}(f_2)B_{2,1} \geq -|B_{2,1}|$. The dominance inequality at x_0 is $|B_{2,1}| \leq \rho_0 \frac{|f_2|}{|f_1|} |B_{1,1}|$, hence

$$\frac{|B_{1,1}|}{|f_1|} - \frac{\text{sign}(f_2)B_{2,1}}{|f_2|} \geq \frac{|B_{1,1}|}{|f_1|} - \frac{|B_{2,1}|}{|f_2|} \geq (1 - \rho_0) \frac{|B_{1,1}|}{|f_1|},$$

so $\partial_t R_{12}(t, x_0) = (1 - \rho_0) \xi_1 K_{\mu_0}(x_0, x_0) (-d_0) \frac{|B_{1,1}|}{|f_1|} + \varepsilon_0(t)$; the leading term is ≥ 0 , hence $\partial_t R_{12}(t, x_0) \geq 0$ whenever it dominates $|\varepsilon_0|$.

At x_1 . The same argument applies by symmetry (indices $1 \leftrightarrow 2$, $x_0 \leftrightarrow x_1$): channel 2 dominates at x_1 and $\partial_t R_{21}(t, x_1) \geq 0$.

Conclusion. The log-ratio R_{12} at x_0 is non-decreasing on I ; strict dominance of channel 1 at x_0 cannot be lost and is amplified whenever $|B_{1,1}|$ is not too small. By symmetry, channel 2 dominates at x_1 .

G. Detailed Proof Sketches

G.1. Well-Posedness: Picard Iteration Details

The key technical result is that the low-rank structure only multiplies constants by r :

Lemma G.1 (Moment bounds for low-rank sums). *For the low-rank sum $H_2(t, c_2; X, W) = \sum_{k=1}^r L_{c_2, k} f_k(t; X, W)$, we have*

$$\mathbb{E}_X \left[\sup_{s \leq t} \sup_{c_2} |H_2(s, c_2; X, W)|^m \right]^{1/m} \leq K \|L\|_{\infty, 1} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} \left[\sup_{s \leq t} |w_1(s, C_1, k)|^m \right]^{1/m},$$

where $\|L\|_{\infty, 1} \leq rK$ under entrywise bounds.

This lemma shows that the forward propagation bounds are multiplied by at most rK , but the structure of the Picard iteration remains unchanged. The contraction mapping argument proceeds as in the full-rank case, with constants depending on $\|L\|_{\infty, 1}$.

H. Quantitative Approximation by the Mean-Field Limit for Low-Rank Networks

This appendix provides a rigorous quantitative bound on the approximation error between finite-width low-rank networks and their mean-field limit. The key result shows that the approximation error scales as $O(1/\sqrt{n_{\min}} + \sqrt{\epsilon})$ with explicit constants, where $n_{\min}$ is the minimum width across layers and ϵ is the learning rate step size. This bound holds for any n_1 and n_2 , independent of the data dimension d , similar to the full-rank case (Nguyen, 2021; Nguyen & Pham, 2023). The bound suggests that widths $n_1, n_2 \approx 1000$ are typically sufficient to observe mean-field behaviors, as empirically validated in (?) for high-dimensional real-world data.

H.1. Main Result: Approximation by the MF Limit

Assumption H.1 (Initialization for Low-Rank Networks). We assume that $\text{ess-sup} \max_{1 \leq k \leq r} |w_1^0(C_1, k)| \leq K$ and $\text{ess-sup} |w_2^0(C_2)| \leq K$, where w_1^0 and w_2^0 are the initial weights as described in the low-rank architecture setup.

Theorem H.2 (Finite-width approximation error bound). *Given a family Init of initialization laws and a tuple $\{n_1, n_2\}$ that is in the index set of Init , perform the coupling procedure for the low-rank architecture as described in Section 3.1. Fix a terminal time $T \in \mathbb{N}_{\geq 0}$. Under Assumptions B.1, H.1, and the low-rank structure with mixing matrix L satisfying $\|L\|_{\infty, 1} \leq rK$, for $\epsilon \leq 1$, we have with probability at least $1 - 2\delta$,*

$$\mathcal{D}_T(W, \mathbf{W}) \leq C_{\text{exp}} \cdot C_{\text{width}} \cdot C_{\text{log}},$$

where $C_{\text{exp}} = e^{K_T(1+rK)}$, $C_{\text{width}} = 1/\sqrt{n_{\min}} + \sqrt{\epsilon}$, $C_{\text{log}} = \sqrt{\log(3(T+1)n_{\max}^2/\delta + e)}$, with $n_{\min} = \min\{n_1, n_2\}$, $n_{\max} = \max\{n_1, n_2\}$, $K_T = K(1+T^K)$, and the factor $(1+rK)$ accounts for the low-rank structure through $\|L\|_{\infty, 1} \leq rK$.

The exponential factor $e^{K_T(1+rK)}$ reflects the channel feature learning structure of low-rank networks: the r channels evolve independently through the mixing matrix L , and the worst-case time complexity becomes exponential in rK when all channels must be learned simultaneously. However, channel feature learning enables a choice of which channels to activate, and in practice, only a subset of channels are needed, avoiding the worst-case exponential scaling.

The theorem gives a connection between $\mathbf{W}(\lfloor t/\epsilon \rfloor)$ (the discrete-time finite-width network) and $W(t)$ (the continuous-time mean-field limit). The key difference from the full-rank case (Nguyen & Pham, 2023) is the multiplicative factor $(1+rK)$ in the exponential constant. In the full-rank setting, the exponential factor is e^{K_T} , while our low-rank architecture introduces an additional $(1+rK)$ factor that accounts for the r independent channels evolving through the mixing matrix L . This factor reflects the channel feature learning structure: the r channels evolve independently through the mixing matrix L , each contributing a factor that reflects the multi-channel nature of the learning dynamics.

Corollary H.3 (Test function approximation quality). *Under the same setting as Theorem H.2, consider any test function $\psi : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ which is K -Lipschitz in the second variable uniformly in the first variable (an example of ψ is the loss $\mathcal{L}$). For any $\delta > 0$, with probability at least $1 - 3\delta$,*

$$\sup_{t \leq T} |\mathbb{E}_Z [\psi(Y, \hat{y}(X; \mathbf{W}(\lfloor t/\epsilon \rfloor)))] - \mathbb{E}_Z [\psi(Y, \hat{y}(X; W(t)))]| \leq e^{2K_T(1+rK)} \left(\frac{1}{\sqrt{n_{\min}}} + \sqrt{\epsilon} \right) \log^{1/2} \left(\frac{3(T+1)n_{\max}^2}{\delta} + e \right),$$

where $\hat{y}(X; \mathbf{W})$ and $\hat{y}(X; W)$ denote the outputs of the finite-width and mean-field networks respectively.

Proof sketch. The proof follows the same structure as in the full-rank case (Nguyen, 2021). Since ψ is K -Lipschitz in the second variable, we have:

$$|\psi(Y, \hat{y}(X; \mathbf{W})) - \psi(Y, \hat{y}(X; W))| \leq K |\hat{y}(X; \mathbf{W}) - \hat{y}(X; W)|.$$

The difference $|\hat{y}(X; \mathbf{W}) - \hat{y}(X; W)|$ can be bounded by the distance $\mathcal{D}_T(W, \mathbf{W})$ using the low-rank structure of the network. Specifically, for the low-rank architecture, the output difference involves:

$$\begin{aligned} |\hat{y}(X; \mathbf{W}) - \hat{y}(X; W)| &\leq \sum_{k=1}^r |L_{C_2, k}| \left| \frac{1}{n_1} \sum_{j_1=1}^{n_1} \mathbf{w}_1(\lfloor t/\epsilon \rfloor, j_1, k) \varphi_1(\langle L^0(C_1(j_1)), X \rangle) \right. \\ &\quad \left. - \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), X \rangle)] \right| + (\text{similar terms for } w_2) \\ &\leq rK \mathcal{D}_T(W, \mathbf{W}) + K \mathcal{D}_T(W, \mathbf{W}) = K(1+rK) \mathcal{D}_T(W, \mathbf{W}), \end{aligned}$$

where we used $\|L\|_{\infty, 1} \leq rK$. Taking expectation over Z and applying Theorem H.2 completes the proof. $\square$

Comparison with the full-rank Case. The main difference between our low-rank result and the full-rank case (Nguyen, 2021) is the appearance of the factor $(1+rK)$ in the exponential constant. In the full-rank case, the bound is:

$$\mathcal{D}_T(W, \mathbf{W}) \leq e^{K_T} \left(\frac{1}{\sqrt{n_{\min}}} + \sqrt{\epsilon} \right) \log^{1/2} \left(\frac{3(T+1)n_{\max}^2}{\delta} + e \right),$$

whereas in our low-rank case, we have:

$$\mathcal{D}_T(W, \mathbf{W}) \leq e^{K_T(1+rK)} \left(\frac{1}{\sqrt{n_{\min}}} + \sqrt{\epsilon} \right) \log^{1/2} \left(\frac{3(T+1)n_{\max}^2}{\delta} + e \right).$$

The factor $(1+rK)$ arises from:

- The low-rank structure of H_2 , which involves a sum over r channels: $H_2(t, c_2; X, W) = \sum_{k=1}^r L_{c_2, k} f_k(t; X, W)$.
- The mixing matrix bounds: $\|L\|_{\infty, 1} = \sup_{c_2} \sum_{k=1}^r |L_{c_2, k}| \leq rK$.
- The backpropagated signal B_k which aggregates over n_2 neurons with mixing coefficients $L_{C_2, k}$.

However, the fundamental structure of the proof remains the same: we still decompose the error into particle coupling and gradient descent discretization, and the scaling with $n_{\min}$ and ϵ is identical. The low-rank structure only affects the exponential constant, not the polynomial scaling. This suggests that the mean-field approximation quality is preserved under low-rank constraints, with the trade-off being a potentially larger (but still finite) exponential constant.

Remark H.4 (Deterministic mixing matrix and hierarchical learning). The mixing matrix L can be chosen deterministically (e.g., on a grid) rather than randomly, since the proof only requires the boundedness condition $\|L\|_{\infty,1} \leq rK$. There is no advantage to maximizing the entries of L ; fixing L deterministically with appropriate structure can enable hierarchical learning, where different channels k specialize to different frequency components or scales through the backpropagated signal B_k . This design choice allows for structured learning dynamics while maintaining the same theoretical guarantees.

H.2. Particle ODEs for Low-Rank Networks

We construct auxiliary trajectories, which we call the *particle ODEs* for the low-rank case. These are continuous-time trajectories of finitely many neurons, averaged over the data distribution, adapted to the low-rank structure:

$$\begin{aligned}\frac{\partial}{\partial t} \tilde{w}_2(t, j_2) &= -\xi_2(t) \mathbb{E}_Z \left[d_L(Z; \tilde{W}(t)) \varphi_2(\tilde{H}_2(t, j_2; X, \tilde{W}(t))) \right], \\ \frac{\partial}{\partial t} \tilde{w}_1(t, j_1, k) &= -\xi_1(t) \mathbb{E}_Z \left[d_L(Z; \tilde{W}(t)) \varphi_1(\langle L^0(C_1(j_1)), X \rangle) \tilde{B}_k(t; X, \tilde{W}(t)) \right],\end{aligned}$$

where $j_1 = 1, \dots, n_1$, $j_2 = 1, \dots, n_2$, $k = 1, \dots, r$, $\tilde{W}(t) = (\tilde{w}_1(t, \cdot, \cdot), \tilde{w}_2(t, \cdot))$, and $t \in \mathbb{R}_{\geq 0}$.

The second-layer output for the particle ODEs is:

$$\tilde{H}_2(t, j_2; X, \tilde{W}) = \sum_{k=1}^r L_{C_2(j_2),k} \frac{1}{n_1} \sum_{j_1=1}^{n_1} \tilde{w}_1(t, j_1, k) \varphi_1(\langle L^0(C_1(j_1)), X \rangle),$$

and the backpropagated signal is:

$$\tilde{B}_k(t; X, \tilde{W}) = \frac{1}{n_2} \sum_{j_2=1}^{n_2} L_{C_2(j_2),k} \varphi'_2(\tilde{H}_2(t, j_2; X, \tilde{W})) \tilde{w}_2(t, j_2).$$

We specify the initialization $\tilde{W}(0)$: $\tilde{w}_1(0, j_1, k) = w_1^0(C_1(j_1), k)$ and $\tilde{w}_2(0, j_2) = w_2^0(C_2(j_2))$. That is, it shares the same initialization with the neural network $\mathbf{W}(0)$, and hence is coupled with the neural network and the MF ODEs.

The existence and uniqueness of the solution to the particle ODEs follows from the same proof as in Theorem 3.1, adapted to account for the low-rank structure. We equip $\tilde{W}(t)$ with the norm:

$$\|\tilde{W}\|_T = \max \left\{ \max_{j_1 \leq n_1, 1 \leq k \leq r} \sup_{t \leq T} |\tilde{w}_1(t, j_1, k)|, \max_{j_2 \leq n_2} \sup_{t \leq T} |\tilde{w}_2(t, j_2)| \right\}.$$

We define the distance measures:

$$\begin{aligned}\mathcal{D}_T(W, \tilde{W}) &= \sup \left\{ |w_1(t, C_1(j_1), k) - \tilde{w}_1(t, C_1(j_1), k)|, |w_2(t, C_2(j_2)) - \tilde{w}_2(t, C_2(j_2))| : \right. \\ &\quad \left. t \leq T, j_1 \leq n_1, j_2 \leq n_2, 1 \leq k \leq r \right\}, \\ \mathcal{D}_T(\tilde{W}, \mathbf{W}) &= \sup \left\{ |\mathbf{w}_1(\lfloor t/\epsilon \rfloor, j_1, k) - \tilde{w}_1(t, C_1(j_1), k)|, \right. \\ &\quad \left. |\mathbf{w}_2(\lfloor t/\epsilon \rfloor, j_2) - \tilde{w}_2(t, C_2(j_2))| : t \leq T, j_1 \leq n_1, j_2 \leq n_2, 1 \leq k \leq r \right\}.\end{aligned}$$

H.3. Key Lemmas

Lemma H.5 (Particle coupling bound). *Under the same setting as Theorem H.2, for any $\delta > 0$, with probability at least $1 - \delta$,*

$$\mathcal{D}_T(W, \tilde{W}) \leq \frac{1}{\sqrt{n_{\min}}} \log^{1/2} \left(\frac{3(T+1)n_{\max}^2}{\delta} + e \right) e^{K_T(1+rK)},$$

in which $n_{\min} = \min\{n_1, n_2\}$, $n_{\max} = \max\{n_1, n_2\}$, and $K_T = K(1 + T^K)$.

Lemma H.6 (Gradient descent discretization error). *Under the same setting as Theorem H.2, for any $\delta > 0$ and $\epsilon \leq 1$, with probability at least $1 - \delta$,*

$$\mathcal{D}_T(\tilde{W}, \mathbf{W}) \leq \sqrt{\epsilon \log \left(\frac{2n_1 n_2 r}{\delta} + e \right)} e^{K_T(1+rK)},$$

in which $K_T = K(1 + T^K)$.

Proof of Theorem H.2. Using the triangle inequality:

$$\mathcal{D}_T(W, \mathbf{W}) \leq \mathcal{D}_T(W, \tilde{W}) + \mathcal{D}_T(\tilde{W}, \mathbf{W}),$$

the result follows immediately from Lemmas H.5 and H.6, noting that the $\log(2n_1 n_2 r / \delta)$ term can be absorbed into the $\log(3(T+1)n_{\max}^2 / \delta + e)$ term up to constants. $\square$

H.4. Proof of Lemma H.5

Proof. In the following, let K_t denote a generic positive constant that may change from line to line and takes the form $K_t = K(1 + t^K)$, such that $K_t \geq 1$ and $K_t \leq K_T$ for all $t \leq T$. We first note that at initialization, $\mathcal{D}_0(W, \tilde{W}) = 0$. Since $\|W\|_0 \leq K$, we have $\|W\|_T \leq K_T$ by the a priori bounds. Furthermore, it is easy to see that $\|\tilde{W}\|_0 \leq \|W\|_0 \leq K$ almost surely. By the same argument, $\|\tilde{W}\|_T \leq K_T$ almost surely.

We decompose the proof into several steps.

Step 1: Main bound with low-rank structure. Let us define, for brevity, the differences specific to the low-rank architecture:

$$\begin{aligned} q_2(t, x, j_2, c_2) &= \tilde{H}_2(t, j_2; x, \tilde{W}(t)) - H_2(t, c_2; x, W(t)), \\ q_{B,k}(t, x) &= \tilde{B}_k(t; x, \tilde{W}(t)) - B_k(t; x, W(t)), \end{aligned}$$

where $H_2(t, c_2; x, W) = \sum_{k=1}^r L_{c_2,k} \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)]$ and $B_k(t; x, W) = \mathbb{E}_{C_2} [L_{C_2,k} \varphi_2'(H_2(t, C_2; x, W)) w_2(t, C_2)]$.

Consider $t \geq 0$. We first bound the difference in the updates between W and $\tilde{W}$.

Bound for w_2 and $\tilde{w}_2$: By Assumption B.1 and the definition of the ODEs:

$$\begin{aligned} & \left| \frac{\partial}{\partial t} \tilde{w}_2(t, j_2) - \frac{\partial}{\partial t} w_2(t, C_2(j_2)) \right| \\ &= \left| \xi_2(t) \mathbb{E}_Z \left[d_L(Z; \tilde{W}(t)) \varphi_2(\tilde{H}_2(t, j_2; X, \tilde{W}(t))) - d_L(Z; W(t)) \varphi_2(H_2(t, C_2(j_2); X, W(t))) \right] \right| \\ &\leq K \mathbb{E}_Z \left[|d_L(Z; \tilde{W}(t)) - d_L(Z; W(t))| |\varphi_2(\tilde{H}_2(t, j_2; X, \tilde{W}(t)))| \right] \\ &\quad + K \mathbb{E}_Z \left[|d_L(Z; W(t))| |\varphi_2(\tilde{H}_2(t, j_2; X, \tilde{W}(t))) - \varphi_2(H_2(t, C_2(j_2); X, W(t)))| \right] \\ &\leq K_t \mathbb{E}_Z [|q_2(t, X, j_2, C_2(j_2))|] + K_t \mathcal{D}_t(W, \tilde{W}), \end{aligned}$$

where we used that d_L is Lipschitz in W , φ_2 is Lipschitz, and H_2 differences are controlled.

Bound for w_1 and $\tilde{w}_1$: For the low-rank case, we have $k = 1, \dots, r$ channels. By the definition of the ODEs:

$$\begin{aligned} & \left| \frac{\partial}{\partial t} \tilde{w}_1(t, j_1, k) - \frac{\partial}{\partial t} w_1(t, C_1(j_1), k) \right| \\ &= \left| \xi_1(t) \mathbb{E}_Z \left[d_L(Z; \tilde{W}(t)) \varphi_1(\langle L^0(C_1(j_1)), X \rangle) \tilde{B}_k(t; X, \tilde{W}(t)) \right. \right. \\ & \quad \left. \left. - d_L(Z; W(t)) \varphi_1(\langle L^0(C_1(j_1)), X \rangle) B_k(t; X, W(t)) \right] \right| \\ &\leq K \mathbb{E}_Z \left[|d_L(Z; \tilde{W}(t)) - d_L(Z; W(t))| |\tilde{B}_k(t; X, \tilde{W}(t))| \right] \\ & \quad + K \mathbb{E}_Z [|d_L(Z; W(t))| |q_{B,k}(t, X)|] \\ &\leq K_t \mathbb{E}_Z [|q_{B,k}(t, X)|] + K_t \mathcal{D}_t(W, \tilde{W}), \end{aligned}$$

where the expectation over j_2 in $\tilde{B}_k$ will be handled via concentration.

Step 2: Decomposition of q_2 with low-rank structure. The key difference from the full-rank case is that H_2 involves a sum over r channels. We decompose:

$$\begin{aligned} |q_2(t, x, j_2, c_2)| &= \left| \sum_{k=1}^r L_{C_2(j_2),k} \left(\frac{1}{n_1} \sum_{j_1=1}^{n_1} \tilde{w}_1(t, j_1, k) \varphi_1(\langle L^0(C_1(j_1)), x \rangle) \right. \right. \\ & \quad \left. \left. - \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)] \right) \right| \\ &\leq \sum_{k=1}^r |L_{C_2(j_2),k}| \left| \frac{1}{n_1} \sum_{j_1=1}^{n_1} \tilde{w}_1(t, j_1, k) \varphi_1(\langle L^0(C_1(j_1)), x \rangle) \right. \\ & \quad \left. - \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)] \right|. \end{aligned}$$

We further decompose each term in the sum:

$$\begin{aligned} & \left| \frac{1}{n_1} \sum_{j_1=1}^{n_1} \tilde{w}_1(t, j_1, k) \varphi_1(\langle L^0(C_1(j_1)), x \rangle) - \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)] \right| \\ &\leq \max_{j_1 \leq n_1} |\tilde{w}_1(t, j_1, k) \varphi_1(\langle L^0(C_1(j_1)), x \rangle) - w_1(t, C_1(j_1), k) \varphi_1(\langle L^0(C_1(j_1)), x \rangle)| \\ & \quad + \left| \frac{1}{n_1} \sum_{j_1=1}^{n_1} w_1(t, C_1(j_1), k) \varphi_1(\langle L^0(C_1(j_1)), x \rangle) - \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)] \right| \\ &\equiv Q_{2,1,k}(x, j_2) + Q_{2,2,k}(x, j_2). \end{aligned}$$

Therefore:

$$|q_2(t, x, j_2, c_2)| \leq \sum_{k=1}^r |L_{C_2(j_2),k}| (Q_{2,1,k}(x, j_2) + Q_{2,2,k}(x, j_2)) \leq rK \max_{1 \leq k \leq r} (Q_{2,1,k}(x, j_2) + Q_{2,2,k}(x, j_2)),$$

where we used $\|L\|_{\infty,1} \leq rK$.

Step 3: Decomposition of $q_{B,k}$ with low-rank structure. For the backpropagated signal difference:

$$\begin{aligned} |q_{B,k}(t, x)| &= \left| \frac{1}{n_2} \sum_{j_2=1}^{n_2} L_{C_2(j_2),k} \varphi_2'(\tilde{H}_2(t, j_2; x, \tilde{W})) \tilde{w}_2(t, j_2) \right. \\ & \quad \left. - \mathbb{E}_{C_2} [L_{C_2,k} \varphi_2'(H_2(t, C_2; x, W)) w_2(t, C_2)] \right|. \end{aligned}$$

We decompose:

$$\begin{aligned}
 |q_{B,k}(t, x)| &\leq \max_{j_2 \leq n_2} \left| L_{C_2(j_2),k} \varphi'_2(\tilde{H}_2(t, j_2; x, \tilde{W})) \tilde{w}_2(t, j_2) \right. \\
 &\quad \left. - L_{C_2(j_2),k} \varphi'_2(H_2(t, C_2(j_2); x, W)) w_2(t, C_2(j_2)) \right| \\
 &\quad + \left| \frac{1}{n_2} \sum_{j_2=1}^{n_2} L_{C_2(j_2),k} \varphi'_2(H_2(t, C_2(j_2); x, W)) w_2(t, C_2(j_2)) \right. \\
 &\quad \left. - \mathbb{E}_{C_2} [L_{C_2,k} \varphi'_2(H_2(t, C_2; x, W)) w_2(t, C_2)] \right| \\
 &\equiv Q_{B,1,k}(x) + Q_{B,2,k}(x).
 \end{aligned}$$

For $Q_{B,1,k}$, using Assumption B.1 and the fact that $|L_{C_2(j_2),k}| \leq K$:

$$\begin{aligned}
 \mathbb{E}_Z[Q_{B,1,k}(X)] &\leq K \max_{j_2 \leq n_2} (|\tilde{w}_2(t, j_2) - w_2(t, C_2(j_2))| \\
 &\quad + |w_2(t, C_2(j_2))| \mathbb{E}_Z[|\tilde{H}_2(t, j_2; X, \tilde{W}) - H_2(t, C_2(j_2); X, W)|]) \\
 &\leq K_t \left(\mathcal{D}_t(W, \tilde{W}) + \max_{j_2 \leq n_2} \mathbb{E}_Z[|q_2(t, X, j_2, C_2(j_2))|] \right).
 \end{aligned}$$

Step 4: Concentration bounds adapted to low-rank. For $Q_{2,1,k}$, we have:

$$\begin{aligned}
 \max_{j_2 \leq n_2} \mathbb{E}_Z[Q_{2,1,k}(X, j_2)] &\leq K \max_{j_1 \leq n_1, 1 \leq k \leq r} |\tilde{w}_1(t, j_1, k) - w_1(t, C_1(j_1), k)| \\
 &\leq K_t \mathcal{D}_t(W, \tilde{W}).
 \end{aligned}$$

For $Q_{2,2,k}$, we apply concentration. Let us write:

$$Z_{1,k}(x, c_1) = w_1(t, c_1, k) \varphi_1(\langle L^0(c_1), x \rangle).$$

Recall that $\{C_1(j_1)\}_{j_1=1}^{n_1}$ are i.i.d. We have:

$$\mathbb{E}[Z_{1,k}(X, C_1(j_1)) | X] = \mathbb{E}_{C_1}[Z_{1,k}(X, C_1)],$$

and $\{Z_{1,k}(X, C_1(j_1))\}_{j_1=1}^{n_1}$ are independent conditional on X . By Assumption B.1, $|Z_{1,k}(X, C_1(j_1))| \leq K_t$ almost surely. Then by concentration inequalities:

$$\mathbb{P}(\mathbb{E}_Z[Q_{2,2,k}(X, j_2)] \geq K_t \gamma_{2,k}) \leq \frac{1}{\gamma_{2,k}} \exp\left(-\frac{n_1 \gamma_{2,k}^2}{K_t}\right).$$

For $Q_{B,2,k}$, we note that almost surely:

$$|L_{C_2(j_2),k} \varphi'_2(H_2(t, C_2(j_2); x, W)) w_2(t, C_2(j_2))| \leq K |L_{C_2(j_2),k}| \leq K^2,$$

by Assumption B.1. Then by concentration inequalities:

$$\mathbb{P}(\mathbb{E}_Z[Q_{B,2,k}(X)] \geq K_t \gamma_{1,k}) \leq \frac{1}{\gamma_{1,k}} \exp\left(-\frac{n_2 \gamma_{1,k}^2}{K_t}\right).$$

Step 5: Combining bounds with union over k . Taking a union bound over $k = 1, \dots, r$ channels, we obtain that for any $\gamma_1, \gamma_2 > 0$ and $t \geq 0$, the event:

$$\begin{aligned}
 &\max \left\{ \max_{j_2 \leq n_2} \mathbb{E}_Z[|q_2(t, X, j_2, C_2(j_2))|], \right. \\
 &\quad \left. \max_{j_1 \leq n_1, 1 \leq k \leq r} \mathbb{E}_Z[|q_{B,k}(t, X)|] \right\} \\
 &\leq K_t(1 + rK)(\mathcal{D}_t(W, \tilde{W}) + \gamma_1 + \gamma_2),
 \end{aligned}$$

holds with probability at least:

$$1 - \frac{n_1 r}{\gamma_1} \exp\left(-\frac{n_2 \gamma_1^2}{K_t(1+rK)^2}\right) - \frac{n_2 r}{\gamma_2} \exp\left(-\frac{n_1 \gamma_2^2}{K_t(1+rK)^2}\right).$$

Step 6: Gronwall argument. Combining the bounds and taking a union bound over a discrete time grid $t \in \{0, \xi, 2\xi, \dots, \lfloor T/\xi \rfloor \xi\}$ for some $\xi \in (0, 1)$, we obtain:

$$\begin{aligned} & \max \left\{ \max_{j_2 \leq n_2} \left| \frac{\partial}{\partial t} \tilde{w}_2(t, j_2) - \frac{\partial}{\partial t} w_2(t, C_2(j_2)) \right|, \right. \\ & \left. \max_{j_1 \leq n_1, 1 \leq k \leq r} \left| \frac{\partial}{\partial t} \tilde{w}_1(t, j_1, k) - \frac{\partial}{\partial t} w_1(t, C_1(j_1), k) \right| \right\} \\ & \leq K_T(1+rK)(\mathcal{D}_t(W, \tilde{W}) + \gamma_1 + \gamma_2 + \xi), \quad \forall t \in [0, T], \end{aligned}$$

with probability at least:

$$1 - \frac{T+1}{\xi} \left[\frac{n_1 r}{\gamma_1} \exp\left(-\frac{n_2 \gamma_1^2}{K_T(1+rK)^2}\right) + \frac{n_2 r}{\gamma_2} \exp\left(-\frac{n_1 \gamma_2^2}{K_T(1+rK)^2}\right) \right].$$

The above event implies:

$$\mathcal{D}_t(W, \tilde{W}) \leq K_T(1+rK) \int_0^t (\mathcal{D}_s(W, \tilde{W}) + \gamma_1 + \gamma_2 + \xi) ds,$$

and hence by Gronwall's lemma and the fact $\mathcal{D}_0(W, \tilde{W}) = 0$:

$$\mathcal{D}_T(W, \tilde{W}) \leq (\gamma_1 + \gamma_2 + \xi) e^{K_T(1+rK)}.$$

The result follows from choosing:

$$\begin{aligned} \xi &= \frac{1}{\sqrt{n_{\max}}}, \\ \gamma_1 &= \frac{K_T(1+rK)}{\sqrt{n_2}} \log^{1/2} \left(\frac{3(T+1)n_{\max}^2 r}{\delta} + e \right), \\ \gamma_2 &= \frac{K_T(1+rK)}{\sqrt{n_1}} \log^{1/2} \left(\frac{3(T+1)n_{\max}^2 r}{\delta} + e \right). \end{aligned}$$

□

H.5. Proof of Lemma H.6

Proof. The proof follows the same structure as the full-rank case (Nguyen, 2021), with careful adaptations for the low-rank structure. The key difference is that we need to account for the r channels in w_1 and the mixing matrix L in all bounds.

We consider $t \leq T$ for a given terminal time $T \in \epsilon \mathbb{N}_{\geq 0}$. We reuse the notation K_t from the proof of Lemma H.5. Note that $K_t \leq K_T$ for all $t \leq T$. We also note that at initialization, $\mathcal{D}_0(\mathbf{W}, \tilde{W}) = 0$.

For brevity, let us define quantities that relate to the difference in the gradient updates between $\mathbf{W}$ and $\tilde{W}$:

$$\begin{aligned} q_2(k, z, \tilde{z}, j_2) &= d_L(z; \mathbf{W}(k)) \varphi_2(\mathbf{H}_2(k, j_2; x, \mathbf{W}(k))) \\ &\quad - d_L(\tilde{z}; \tilde{W}(k\epsilon)) \varphi_2(\tilde{H}_2(k\epsilon, j_2; \tilde{x}, \tilde{W}(k\epsilon))), \\ r_2(k, z, j_2) &= \xi_2(k\epsilon) d_L(z; \tilde{W}(k\epsilon)) \varphi_2(\tilde{H}_2(k\epsilon, j_2; x, \tilde{W}(k\epsilon))) \\ &\quad - \xi_2(k\epsilon) \mathbb{E}_Z[d_L(Z; \tilde{W}(k\epsilon)) \varphi_2(\tilde{H}_2(k\epsilon, j_2; X, \tilde{W}(k\epsilon)))], \\ q_1(k, z, \tilde{z}, j_1, k') &= d_L(z; \mathbf{W}(k)) \varphi_1(\langle L^0(C_1(j_1)), x \rangle) \mathbf{B}_{k'}(k; x, \mathbf{W}(k)) \\ &\quad - d_L(\tilde{z}; \tilde{W}(k\epsilon)) \varphi_1(\langle L^0(C_1(j_1)), \tilde{x} \rangle) \tilde{B}_{k'}(k\epsilon; \tilde{x}, \tilde{W}(k\epsilon)), \\ r_1(k, z, j_1, k') &= \xi_1(k\epsilon) d_L(z; \tilde{W}(k\epsilon)) \varphi_1(\langle L^0(C_1(j_1)), x \rangle) \tilde{B}_{k'}(k\epsilon; x, \tilde{W}(k\epsilon)) \\ &\quad - \xi_1(k\epsilon) \mathbb{E}_Z[d_L(Z; \tilde{W}(k\epsilon)) \varphi_1(\langle L^0(C_1(j_1)), X \rangle) \tilde{B}_{k'}(k\epsilon; X, \tilde{W}(k\epsilon))], \end{aligned}$$

where $\mathbf{B}_{k'}(k; x, \mathbf{W}) = \frac{1}{n_2} \sum_{j_2=1}^{n_2} L_{C_2(j_2), k'} \varphi'_2(\mathbf{H}_2(k, j_2; x, \mathbf{W})) \mathbf{w}_2(k, j_2)$.

By time-interpolation estimates (similar to Claim 1 in the full-rank case) and Assumption B.1, we have:

$$\begin{aligned} |\mathbf{w}_2(\lfloor t/\epsilon \rfloor, j_2) - \tilde{w}_2(t, j_2)| &\leq K \max_{j_2 \leq n_2} [Q_{2,1}(\lfloor t/\epsilon \rfloor, j_2) + Q_{2,2}(\lfloor t/\epsilon \rfloor, j_2)] + tK_t\epsilon, \\ |\mathbf{w}_1(\lfloor t/\epsilon \rfloor, j_1, k) - \tilde{w}_1(t, j_1, k)| &\leq K \max_{j_1 \leq n_1, 1 \leq k \leq r} [Q_{1,1}(\lfloor t/\epsilon \rfloor, j_1, k) + Q_{1,2}(\lfloor t/\epsilon \rfloor, j_1, k)] + tK_t\epsilon, \end{aligned}$$

where:

$$\begin{aligned} Q_{2,1}(\lfloor t/\epsilon \rfloor, j_2) &= \epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} |q_2(\ell, z(\ell), z(\ell), j_2)|, \\ Q_{2,2}(\lfloor t/\epsilon \rfloor, j_2) &= \left| \epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} r_2(\ell, z(\ell), j_2) \right|, \\ Q_{1,1}(\lfloor t/\epsilon \rfloor, j_1, k) &= \epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} |q_1(\ell, z(\ell), z(\ell), j_1, k)|, \\ Q_{1,2}(\lfloor t/\epsilon \rfloor, j_1, k) &= \left| \epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} r_1(\ell, z(\ell), j_1, k) \right|. \end{aligned}$$

Bounding the terms: For $Q_{2,1}$, using Assumption B.1 and the low-rank structure:

$$\begin{aligned} |q_2(\ell, z(\ell), z(\ell), j_2)| &\leq K |d_L(z(\ell); \mathbf{W}(\ell)) - d_L(z(\ell); \tilde{W}(\ell\epsilon))| \\ &\quad + K |d_L(z(\ell); \tilde{W}(\ell\epsilon))| |\mathbf{H}_2(\ell, j_2; x(\ell), \mathbf{W}(\ell)) - \tilde{H}_2(\ell\epsilon, j_2; x(\ell), \tilde{W}(\ell\epsilon))| \\ &\leq K_t \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}) + K_t \sum_{k=1}^r |L_{C_2(j_2), k}| \max_{j_1 \leq n_1} |\mathbf{w}_1(\ell, j_1, k) - \tilde{w}_1(\ell\epsilon, j_1, k)| \\ &\leq K_t(1 + rK) \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}), \end{aligned}$$

which yields:

$$\max_{j_2 \leq n_2} Q_{2,1}(\lfloor t/\epsilon \rfloor, j_2) \leq K_t(1 + rK) \epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}).$$

For $Q_{1,1}$, similarly:

$$\begin{aligned} |q_1(\ell, z(\ell), z(\ell), j_1, k)| &\leq K |d_L(z(\ell); \mathbf{W}(\ell)) - d_L(z(\ell); \tilde{W}(\ell\epsilon))| |\mathbf{B}_k(\ell; x(\ell), \mathbf{W}(\ell))| \\ &\quad + K |d_L(z(\ell); \tilde{W}(\ell\epsilon))| |\mathbf{B}_k(\ell; x(\ell), \mathbf{W}(\ell)) - \tilde{B}_k(\ell\epsilon; x(\ell), \tilde{W}(\ell\epsilon))| \\ &\leq K_t(1 + rK) \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}), \end{aligned}$$

which yields:

$$\max_{j_1 \leq n_1, 1 \leq k \leq r} Q_{1,1}(\lfloor t/\epsilon \rfloor, j_1, k) \leq K_t(1 + rK) \epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}).$$

For $Q_{2,2}$ and $Q_{1,2}$, we use martingale concentration. The martingale differences are bounded: $|r_2(\ell, z(\ell), j_2)| \leq K_t$ and $|r_1(\ell, z(\ell), j_1, k)| \leq K_t(1 + rK)$ almost surely by Assumption B.1 and the low-rank structure. Then by martingale concentration inequalities:

$$\begin{aligned} \mathbb{P} \left(\max_{j_2 \leq n_2} \max_{\ell \in \{0, 1, \dots, T/\epsilon\}} Q_{2,2}(\ell, j_2) \geq \xi \right) &\leq 2n_2 \exp \left(-\frac{\xi^2}{K_T(T+1)\epsilon} \right); \\ \mathbb{P} \left(\max_{j_1 \leq n_1, 1 \leq k \leq r} \max_{\ell \in \{0, 1, \dots, T/\epsilon\}} Q_{1,2}(\ell, j_1, k) \geq \xi \right) &\leq 2n_1 r \exp \left(-\frac{\xi^2}{K_T(1+rK)^2(T+1)\epsilon} \right). \end{aligned}$$

Putting everything together: All the above results give us:

$$\mathcal{D}_{\lfloor t/\epsilon \rfloor \epsilon}(\tilde{W}, \mathbf{W}) \leq K_T(1+rK)\epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}) + \xi + TK_T\epsilon, \quad \forall t \leq T,$$

which holds with probability at least:

$$1 - 2n_1r \exp\left(-\frac{\xi^2}{K_T(1+rK)^2(T+1)\epsilon}\right) - 2n_2 \exp\left(-\frac{\xi^2}{K_T(T+1)\epsilon}\right).$$

The above event implies, by Gronwall's lemma:

$$\mathcal{D}_T(\tilde{W}, \mathbf{W}) \leq (\xi + \epsilon)e^{K_T(1+rK)}.$$

Choosing $\xi = K_T(1+rK)\sqrt{(T+1)\epsilon \log(2n_1n_2r/\delta)}$ completes the proof. $\square$

H.6. Discussion: Width Requirement and Exponential Factor

The bound in Theorem H.2 shows that the approximation error scales as:

$$\mathcal{D}_T(W, \mathbf{W}) \leq e^{K_T(1+rK)} \left(\frac{1}{\sqrt{n_{\min}}} + \sqrt{\epsilon} \right) \log^{1/2} \left(\frac{3(T+1)n_{\max}^2}{\delta} + e \right).$$

For typical values $T \leq 10$, $K \approx 1$, $r \leq 100$, and $\epsilon \approx 0.001$, we have $K_T(1+rK) \leq 1000$ (roughly), so $e^{K_T(1+rK)} \leq e^{1000}$, which is an extremely large exponential factor. This reflects the worst-case scenario where all r channels must be learned simultaneously. However, in practice, channel specialization enables a more favorable learning regime: channels progressively capture different frequency components, avoiding the worst-case exponential scaling. The actual convergence rate is determined by the favorable loss landscape structure (see Section 6) rather than this worst-case bound.

The key qualitative insight is that the error *decreases* as $n_{\min}$ increases, with rate $O(1/\sqrt{n_{\min}})$ independent of data dimension d . The exponential factor $e^{K_T(1+rK)}$ reflects the multi-channel structure but does not dominate in practice due to channel specialization.

Empirical validation in (?) confirms that networks with width ≈ 1000 trained on high-dimensional real-world data (e.g., $d \approx 1000$) exhibit mean-field behaviors, supporting our theoretical prediction that the required width is independent of d .

I. Additional Experimental Details

I.1. Ablation Studies on Frequency-Dependent Function Approximation

Our ablation studies demonstrate that freezing mixing consistently improves performance across all frequency regimes compared to fully trainable networks. Moreover, low-rank configurations achieve excellent performance, with optimal rank depending on the frequency regime. Detailed results are provided in Table 2 below.

Key findings: (1) **fixWb=True is critical:** At high frequencies (144, 48), fixWb=False with rank 25 fails catastrophically (error 0.791), while fixWb=True achieves excellent performance (error 0.0027). (2) **Optimal rank depends on frequency:** Rank 20-25 provides best balance across frequencies, with rank 20 optimal at high frequencies. (3) **fixWb improves mean performance:** fixWb=True achieves 3.7 $\times$ better mean test error and 2.5 $\times$ lower variance across all configurations. (4) **Frequency scaling:** Error increases exponentially with frequency (36 $\rightarrow$ 72: 3 $\times$, 72 $\rightarrow$ 144: 39 $\times$), demonstrating the non-convex optimization challenge.

Efficiency ratio analysis: The efficiency ratio (Test MSE (False) / Test MSE (True)) quantifies the improvement from using frozen random features. At medium frequencies (72, 24), the efficiency ratio ranges from 0.3 to 22,941, with rank 100 achieving a dramatic 22,941 $\times$ improvement. At high frequencies (144, 48), the efficiency ratio reaches up to 52,661 $\times$ for rank 100, demonstrating that frozen random features become increasingly critical as frequency increases. The efficiency ratio exhibits strong rank dependence: lower ranks (10-25) show moderate improvements (3-31 $\times$), while

Table 2. Ablation study results on frequency-dependent function approximation. `fixWb=True` (frozen random features) consistently outperforms `fixWb=False` (fully trainable), and low-rank configurations achieve excellent performance across all frequency regimes. Params shows parameter count (K=thousands, M=millions). Ratio = Test MSE (False) / Test MSE (True) for same rank, showing improvement factor. Low-rank (rank 10-100) reduces parameters by 77-97% compared to full-rank (1024), while achieving superior performance through channel feature learning that enables hierarchical frequency learning.

Frequency	fixWb	Rank	Params	Test MSE	Train MSE	Ratio
	False	10	62K	2.8e-02	2.7e-02	–
	True	10	31K	8.9e-04	9.5e-04	31
	False	15	93K	1.8e-03	1.5e-03	–
	True	15	47K	2.1e-03	3.0e-03	0.9
	False	20	124K	5.7e-04	9.2e-04	–
	True	20	62K	1.8e-03	1.8e-03	0.3
(36, 12)	False	25	155K	1.6e-02	1.7e-02	–
	True	25	78K	8.0e-04	6.4e-04	20
	False	50	311K	1.2e-03	7.5e-04	–
	True	50	155K	1.9e-03	1.4e-03	0.6
	False	100	622K	7.8e-01	7.8e-01	–
	True	100	311K	3.4e-05	3.1e-09	22941
(72, 24)	False	1024	4.8M	7.8e-01	7.8e-01	–
	True	1024	2.4M	7.8e-01	7.8e-01	1.0
	False	10	62K	2.3e-01	2.3e-01	–
	True	10	31K	3.2e-01	3.1e-01	0.7
	False	15	93K	1.4e-01	1.4e-01	–
	True	15	47K	3.9e-02	3.8e-02	3.6
	False	20	124K	1.9e-01	1.8e-01	–
	True	20	62K	6.4e-04	1.0e-03	299
(144, 48)	False	25	155K	7.9e-01	7.9e-01	–
	True	25	78K	2.7e-03	2.9e-03	289
	False	100	622K	7.9e-01	7.9e-01	–
	True	100	311K	1.5e-05	1.8e-06	52661
(144, 48)	–	1024	4.8M	N/A	N/A	–

higher ranks (100) achieve orders-of-magnitude improvements (22,941-52,661×). This pattern aligns with our theoretical analysis: frozen random features maintain universal approximation throughout training, enabling channel feature learning that becomes more effective at higher ranks where more channels can learn different spatial-frequency features. The few cases where the efficiency ratio is less than 1 (e.g., rank 15 at (72, 24) with ratio 0.9, rank 20 at (72, 24) with ratio 0.3) occur when the fully trainable network happens to find a favorable initialization, but these are exceptions rather than the rule.

I.2. Dataset Descriptions

We evaluate our architecture on frequency-dependent function approximation tasks, which provide a controlled setting for studying how low-rank networks learn multi-scale features. The target function is:

$$f(x) = \cos(f_1 \pi x^2) - 0.8 \cos(f_2 \pi x^2), \quad x \in [-1, 1],$$

where (f_1, f_2) are frequency pairs that determine the complexity of the function. It is symmetric about $x = 0$ (both terms use x^2 , so $f(-x) = f(x)$). The frequency pairs are as in `experiments/former/SinQuad/benchmark.py` (base pair (36, 12); we also use scaled regimes (72, 24) and (144, 48)). This task is inherently non-convex and tests the network’s capacity to learn high-frequency components while maintaining convergence to a global minimizer.

We test three frequency regimes that progressively increase in difficulty:

- **Low frequency** (36, 12): 1000 training samples, 1234 test samples
- **Medium frequency** (72, 24): 2000 training samples, 2468 test samples
- **High frequency** (144, 48): 4000 training samples, 4936 test samples

The training data is generated using a uniform grid on $[-1, 1]$, while the test set uses random samples from a different distribution to assess generalization. The number of training samples scales linearly with frequency to maintain consistent sampling density relative to the function’s Nyquist frequency. This scaling ensures that higher-frequency functions receive proportionally more training data, though the exponential difficulty increase with frequency (see Section 5) suggests that even this scaling may be insufficient for the highest frequencies.

I.3. Hyperparameter Settings

We conduct a comprehensive ablation study across multiple hyperparameters to understand the sensitivity of low-rank networks to different configurations. Table 3 summarizes all hyperparameter settings used in our experiments.

Table 3. Complete hyperparameter settings for all experiments. The main ablation study (reported in Table 2) uses batch size 100, while additional experiments explore batch sizes 50 and 200.

Hyperparameter	Values
Architecture	
Depth L	8 layers
Width n	1024 neurons per layer
Activation	ReLU
Rank r	{10, 15, 20, 25, 50, 100, 1024}
fixWb	True (frozen), False (trainable)
Optimization	
Optimizer	Adam
Learning rate	0.001 (initial)
Scheduler	StepLR ($\gamma = 0.9$, step_size=100)
Gradient clipping	max_norm=1.0
Batch size	{50, 100, 200}
Training epochs	10,000 (all runs)
Data	
Frequency pairs (f_1, f_2)	(36, 12), (72, 24), (144, 48)
Training samples	1000, 2000, 4000 (by frequency)
Test samples	1234, 2468, 4936 (by frequency)
Input domain	$x \in [-1, 1]$
Training Details	
Random seed	42
Device	CUDA (GPU)
Checkpointing	Every 500 epochs
Evaluation frequency	Every 50 epochs

The total experimental space consists of 90 configurations: 6 rank values $\times$ 2 fixWb settings $\times$ 3 frequency pairs $\times$ 3 batch sizes (with some configurations omitted for rank 100 and full-rank 1024). All runs complete 10,000 epochs without early stopping to ensure fair comparison and to study convergence behavior over extended training.

I.4. Additional Plots

Beyond the main results presented in Section 5, we generate extensive visualizations for all 90 configurations. These additional plots provide deeper insights into the internal dynamics of low-rank networks:

- **Loss evolution plots:** Training loss over all 10,000 epochs for each configuration, showing convergence patterns and stability across different ranks and fixWb settings.
- **Error evolution plots:** Train and test error trajectories, revealing generalization behavior and the train-test gap over time.
- **Final prediction plots:** True vs. learned function comparisons for all configurations, demonstrating the quality of function approximation across frequency regimes.
- **Layer-wise partial function visualizations:** For each configuration, we visualize up to 36 channels per layer, showing how the network builds hierarchical representations through the 8-layer architecture. These plots reveal how different channels specialize to different frequency components, providing empirical validation of the channel specialization mechanism described in Section 4.

All visualizations are saved in the experiment results directories and can be used to analyze the relationship between rank, fixWb, frequency, and the learned representations. The partial function visualizations are particularly valuable for understanding how channel specialization manifests in practice, showing how channels progressively capture higher frequencies as training progresses.

I.5. Implementation Details

I.6. Experimental Setup

Table 4. Hyperparameters: ablation (Table 2) and mean-field analysis (Figure 6).

Config	L	n	r	Target	fixWb	Optim	lr / epochs / batch
Ablation	8	1024	10-50	$(36, 12), (72, 24), (144, 48)$	T/F	Adam	0.001 / 10k / 100
Mean-field	3	1024	15	$\cos(8\pi x)$	–	SGD	0.01 / 10k / 160

Ablation: $f(x) = \cos(f_1\pi x^2) - 0.8\cos(f_2\pi x^2)$ on $[-1, 1]$, step decay $\gamma = 0.9$, max_norm=1, uniform init. $[-1, 1]/\sqrt{n}$; log-ratios $R_{i,j}$ at $x = 0$.